

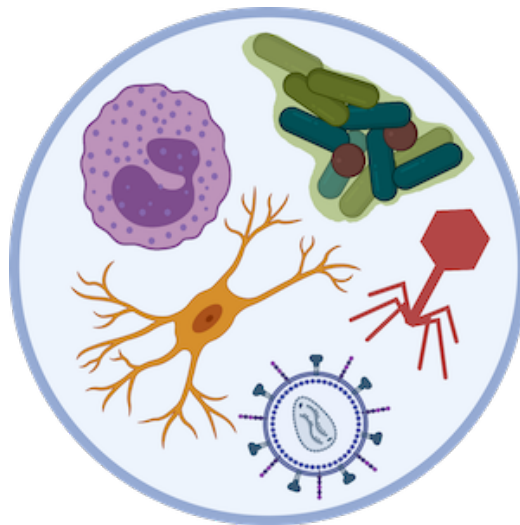
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# **IGCSE Double Award Science Biology Revised June 2026**

## **Unit 1 Cells & Microbes Student Booklet**



Lesson 1: Characteristics of Living Things

Lesson 2: Animal and Plant Cells

Lesson 3: Diffusion

Lesson 4 & 5: Osmosis

Lesson 6: Active Transport

Lesson 7: Pathogens

Lesson 8: Revision Resources



## **Double Award Biology - Cells and Microbes Syllabus Statements**

### **1      The nature and variety of living organisms**

#### **(a)      Characteristics of living organisms**

##### **Students should:**

1.1      understand how living organisms share the following characteristics:

- they require nutrition
- they respire
- they excrete their waste
- they respond to their surroundings
- they move
- they control their internal conditions
- they reproduce
- they grow and develop.

#### **(b)      Variety of living organisms**

##### **Students should:**

1.2      describe the common features shown by eukaryotic organisms: plants, animals, fungi and protocists

Plants: these are multicellular organisms; their cells contain chloroplasts and are able to carry out photosynthesis; their cells have cellulose cell walls; they store carbohydrates as starch or sucrose. Examples include flowering plants, such as a cereal (for example, maize), and a herbaceous legume (for example, peas or beans).

Animals: these are multicellular organisms; their cells do not contain chloroplasts and are not able to carry out photosynthesis; they have no cell walls; they usually have nervous co-ordination and are able to move from one place to another; they often store carbohydrate as glycogen. Examples include mammals (for example, humans) and insects (for example, housefly and mosquito).

Fungi: these are organisms that are not able to carry out photosynthesis; their body is usually organised into a mycelium made from thread-like structures called hyphae, which contain many nuclei; some examples are single-celled; their cells have walls made of chitin; they feed by extracellular secretion of digestive enzymes onto food material and absorption of the organic products; this is known as saprotrophic nutrition; they may store carbohydrate as glycogen. Examples include *Mucor*, which has the typical fungal hyphal structure, and yeast, which is single-celled.

Protoctists: these are microscopic single-celled organisms. Some, like *Amoeba*, that live in pond water, have features like an animal cell, while others, like *Chlorella*, have chloroplasts and are more like plants. A pathogenic example is *Plasmodium*, responsible for causing malaria.

##### **Students should:**

1.3      describe the common features shown by prokaryotic organisms such as bacteria

Bacteria: these are microscopic single-celled organisms; they have a cell wall, cell membrane, cytoplasm and plasmids; they lack a nucleus but contain a circular chromosome of DNA; some bacteria can carry out photosynthesis but most feed off other living or dead organisms. Examples include *Lactobacillus bulgaricus*, a rod-shaped bacterium used in the production of yoghurt from milk, and

*Pneumococcus*, a spherical bacterium that acts as the pathogen causing pneumonia.

## **Double Award Biology - Cells and Microbes Syllabus Statements**

- 1.4 understand the term pathogen and know that pathogens may include fungi, bacteria, protoctists or viruses
- Viruses: these are not living organisms. They are small particles, smaller than bacteria; they are parasitic and can reproduce only inside living cells; they infect every type of living organism. They have a wide variety of shapes and sizes; they have no cellular structure but have a protein coat and contain one type of nucleic acid, either DNA or RNA. Examples include the tobacco mosaic virus that causes discolouring of the leaves of tobacco plants by preventing the formation of chloroplasts, the influenza virus that causes 'flu' and the HIV virus that causes AIDS.

## **2 Structure and functions in living organisms**

### **(a) Level of organisation**

**Students should:**

- 2.1 describe the levels of organisation in organisms: organelles, cells, tissues, organs and systems

### **(b) Cell structure**

**Students should:**

- 2.2 describe cell structures, including the nucleus, cytoplasm, cell membrane, cell wall, mitochondria, chloroplasts, ribosomes and vacuole
- 2.3 describe the functions of the nucleus, cytoplasm, cell membrane, cell wall, mitochondria, chloroplasts, ribosomes and vacuole
- 2.4 know the similarities and differences in the structure of plant and animal cells

### **(d) Movement of substances into and out of cells**

**Students should:**

- 2.15 understand the processes of diffusion, osmosis and active transport by which substances move into and out of cells
- 2.16 understand how factors affect the rate of movement of substances into and out of cells, including the effects of surface area to volume ratio, distance, temperature and concentration gradient
- 2.17 *practical: investigate diffusion and osmosis using living and non-living systems*

### **(h) Transport**

**Students should:**

- 2.51 understand why simple, unicellular organisms can rely on diffusion for movement of substances in and out of the cell

## Cells and Microbes – Glossary of Key Words

| Keyword                              | Definition                                                                                                                                                                                                       |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Active Transport<br>Active Transport | The movement of substances from a more dilute solution to a more concentrated solution (against a concentration gradient). This requires the use of energy from respiration.                                     |
| ATP<br>ATP                           | ATP (adenosine triphosphate) is the energy-carrying molecule used in cells because it can release energy very quickly. Energy is released from ATP when the end phosphate is removed.                            |
| Bacteria<br>Bacteria                 | Microscopic single-celled organisms. They may produce toxins that damage tissues and make us feel ill. Examples include <i>Salmonella</i> , which causes food poisoning, and the pathogen that causes gonorrhea. |
| Cell<br>Cell                         | The basic building block of all living organisms.                                                                                                                                                                |
| Cell (Plasma) Membrane<br>Membrane   | A partially permeable membrane that surrounds the cytoplasm and controls the movement of substances into and out of the cell.                                                                                    |
| Cell Wall<br>Cell Wall               | A rigid structure found in plant cells and bacterial cells that strengthens the cell and provides support.                                                                                                       |
| Cellulose<br>Cellulose               | The main structural component in the cell walls of plant cells.                                                                                                                                                  |
| Chlorophyll<br>Chlorophyll           | A green pigment found in chloroplasts which absorbs light for photosynthesis.                                                                                                                                    |
| Chloroplast<br>Chloroplast           | An organelle that contains the pigment chlorophyll which absorbs light so the plant can carry out photosynthesis.                                                                                                |
| Concentrated Solution<br>Solution    | A solution that contains a large amount of solute.                                                                                                                                                               |
| Concentration<br>Concentration       | A measure of the amount of solute in a solution.                                                                                                                                                                 |
| Concentration Gradient<br>Gradient   | The difference in concentration between two solutions on either side of a membrane.                                                                                                                              |
| Cytoplasm<br>Cytoplasm               | The jelly-like substance in which sub-cellular structures are found and where most chemical reactions occur.                                                                                                     |
| Diffusion<br>Diffusion               | The net movement of particles from an area of higher concentration to an area of lower concentration (down the concentration gradient).                                                                          |
| Dilute Solution<br>Dilute Solution   | A solution that contains a small amount of solute.                                                                                                                                                               |
| Electron Microscope<br>Microscope    | A microscope that uses a beam of electrons to produce a magnified image of an object. This type of microscope has high magnification and resolution.                                                             |
| Eukaryotic Cell<br>Eukaryotic Cell   | A type of cell that has a nucleus.                                                                                                                                                                               |
| Flagellum<br>Flagellum               | A tail-like structure that enables bacteria and sperm cells to move.                                                                                                                                             |
| Fungi<br>Fungi                       | A large group of eukaryotic organisms that contain single-celled yeasts, moulds and mushrooms.                                                                                                                   |

|                                |                                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Light Microscope               | A microscope that uses lenses and visible light to produce a magnified image of an object. This type of microscope has low magnification and resolution. |
| Magnification                  | The amount that an image of something is scaled up when viewed through a microscope. Magnification = image size ÷ actual size                            |
| Mitochondria                   | Structures in the cytoplasm of all cells where aerobic respiration takes place (singular is mitochondrion).                                              |
| Nucleus                        | An organelle that contains the genetic material of the cell and controls the activities of the cell.                                                     |
| Organ                          | A group of tissues that work together to perform a specific function.                                                                                    |
| Organ System                   | A group of organs working together to perform a function.                                                                                                |
| Organelle                      | A structure found within a cell. Also called a sub-cellular structure.                                                                                   |
| Organism                       | A single living thing. Can be multicellular or single-celled.                                                                                            |
| Osmosis                        | The diffusion of water from a dilute solution to a concentrated solution through a partially permeable membrane.                                         |
| Pathogen                       | A microorganism that causes infectious disease.                                                                                                          |
| Prokaryotic Cell               | A simple cell that does not have a nucleus – the DNA is free in the cytoplasm.                                                                           |
| Protoctist                     | A eukaryotic, usually single-celled organism of the kingdom Protista.                                                                                    |
| Ribosome                       | A sub-cellular structure that carries out protein synthesis using the genetic code from DNA.                                                             |
| Selectively Permeable Membrane | A membrane that allows small molecules (e.g. water and certain solutes) to pass across it but does not allow the passage of large molecules.             |
| Surface Area To Volume Ratio   | The internal volume of the object compared with the amount of area where it contacts its environment.                                                    |
| Tissue                         | A group of cells with a similar structure and function working together.                                                                                 |
| Vacuole                        | An organelle filled with sap to keep the cell rigid enough to support the plant.                                                                         |
| Virus                          | An ultramicroscopic infectious non-cellular organism that can replicate inside the cells of living hosts, with negative consequences.                    |

## Lesson 1: Characteristics of Living Organisms

What makes something alive? State what is meant by the acronym MRS C GREN.

M ~~Movement~~ -- living things move.....

R ~~Respiration~~ -- living things respire to release energy from food.....

S ~~Sensitivity~~ -- living things respond to their surroundings.....

C ~~Cells~~ -- living things are made of cells.....

G ~~Growth~~ -- living things grow and develop.....

R ~~Reproduction~~ -- living things reproduce.....

E ~~Excretion~~ -- living things excrete their waste.....

N ~~Nutrition~~ -- living things require nutrition (need food).....

✓ True ~~or~~ False?

1. All living organisms have eyes.
2. All living organisms breathe.
3. All living organisms need nutrition.
4. All living organisms can reproduce.
5. All living organisms do digestion.

Exam Practice

Q1. All living organisms have certain characteristics.

(a) Describe what is meant by the following characteristics:

(i) respiration

(2)

.....

.....

.....

(ii) control of their internal conditions

(2)

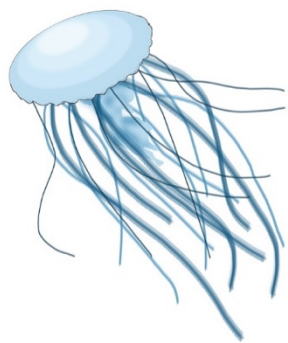
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.....

Respiration is the release of energy from food molecules such as glucose, by a series of chemical reactions in cells. It happens in all living cells.

Keeping the internal conditions of the body (e.g. temperature, water content, blood glucose) within the limits needed for enzymes to work, despite changes inside and outside the body.



There are over 2000 types of jellyfish around the world. Jellyfish swim by rhythmic pulsations of the umbrella. The tentacles on a jellyfish are covered with millions of stinging cells. Jellyfish usually feed on small fish that get caught in their tentacles or sometimes even other jellyfish.

Jellyfish take oxygen into their cells directly from the water around them. Jellyfish don't actually have brains but they do have something called a nerve net which can coordinate responses. Most Jellyfish release sperm or eggs directly into the water around them. Jellyfish have no blood, bones or hearts. But some of them can do incredible things like glow in the dark.

Jellyfish are some of the oldest living creatures on Earth and have probably been around on the planet for over 600 million years. Jellyfish numbers are increasing as they are one of the few living organisms that are thriving with warmer more acidic oceans.

Q2. Suggest and explain 3 pieces of evidence from the passage that show that jellyfish are a type of living organism.

(3)

Any 3, each explained: reproduction (numbers increasing), nutrition and growth (they feed and..... grow), respiration, excretion of waste, movement (response to stimuli).....

Q3. Suggest how jellyfish carry out excretion.

(1)

Evidence jellyfish are living: they reproduce (numbers increase); they feed on other organisms and grow; they respire and excrete waste, and they respond to stimuli by moving.

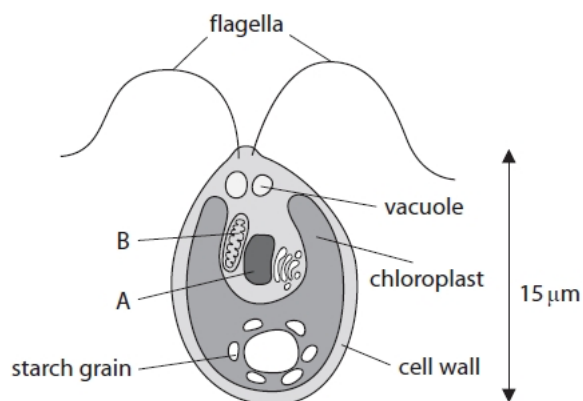
Q4: Some jellyfish can clone themselves to make identical copies. Suggest how this also shows they are living organisms.

(1)

Cloning makes new identical copies of the organism, which is reproduction---a characteristic of all living organisms. They move (swim by pulsations), respire (take oxygen from the water), respond to their surroundings (nerve net coordinates responses), reproduce (release sperm or eggs) and feed on fish (nutrition). They excrete their waste products (e.g. carbon dioxide) by diffusion out of their cells through their body surface into the surrounding water.

| Kingdom          | Characteristics                                              | Examples                                                                                                  | Cell Diagram                                                                                                                 |
|------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Animal</b>    |                                                              |                                                                                                           |                                                                                                                              |
| <b>Plant</b>     | Mammals, e.g. humans; insects, e.g. housefly and mosquito    | Animal cell: cell membrane, cytoplasm, nucleus, mitochondria and ribosomes; no cell wall, no chloroplasts | Plant cell: cellulose cell wall, cell membrane, cytoplasm, nucleus, chloroplasts, large sap vacuole, mitochondria, ribosomes |
| <b>Fungi</b>     | Maize (a cereal) and a herbaceous legume, e.g. peas or beans | Mucor (has the typical fungal hyphal structure) and yeast (single-celled)                                 | Fungal cell (e.g. yeast): cell wall made of chitin, cell membrane, cytoplasm, nucleus, mitochondria; no chloroplasts         |
| <b>Bacteria</b>  |                                                              |                                                                                                           |                                                                                                                              |
| <b>Protocist</b> |                                                              |                                                                                                           |                                                                                                                              |

**Q1.** The diagram shows *Chlamydomonas*, a single-celled protist that is found in water.



(a) Name the structures labelled A and B.

(2)

A **chloroplast**.....

B **nucleus**.....

(b) (i) *Chlamydomonas* has some structures that are normally found in plant cells, but not found in animal cells.

Name two of these structures.

(2)

1 **chloroplast**.....

2 **cell wall**.....

(ii) State why *Chlamydomonas* is not classified as a plant.

(1)

.....  
**It is a single-celled (unicellular) organism, whereas plants are multicellular organisms.**

(c) Explain how *Chlamydomonas* obtains energy.

(2)

**Chlamydomonas has chloroplasts, so it makes its own food by photosynthesis using light energy; it then  
 respire to release the energy it needs.**

(d) *Chlamydomonas* has flagella.

Suggest a function of these flagella.

(1)

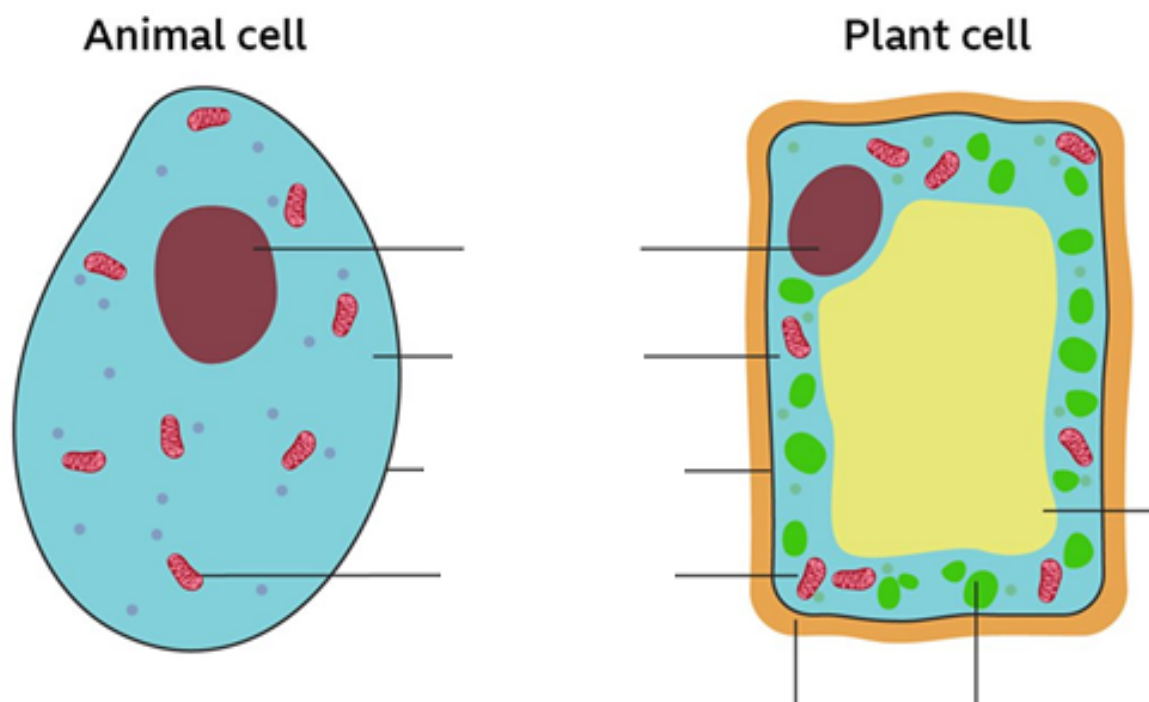
.....  
**The flagella beat (whip) to move the organism through the water - they are used for movement/locomotion.**

· *Chlamydomonas* diagram: single-celled protist with a cup-shaped chloroplast, nucleus, eyespot and two flagella. A = chloroplast, B = nucleus.

It contains chloroplasts with chlorophyll, so it absorbs light energy and photosynthesises to make carbohydrate; energy is then released by respiration.

## Lesson 2: Animal and Plant Cells

Label the cell structures:



Match the cell structure to its function:

| Cell Structure                                             |
|------------------------------------------------------------|
| Cell Wall - provides strength and support for the cell     |
| Vacuole - stores water and nutrients                       |
| Cell Membrane - controls what comes in and out of the cell |
| Chloroplast - where photosynthesis happens                 |
| Cytoplasm - where chemical reactions happen                |
| Nucleus - holds the DNA                                    |
| Mitochondria - where energy is released from food          |

| Function                                   |
|--------------------------------------------|
| Stores water and nutrients                 |
| Controls what comes in and out of the cell |
| Holds the DNA                              |
| Where photosynthesis happens               |
| Provides strength and support for the cell |
| Where energy is released from food         |
| Where chemical reactions happen            |

What does organelle mean?

A structure found within a cell (also called a sub-cellular structure).

Which organelles are found in both animal and plant cells?

Nucleus, cytoplasm, cell membrane, mitochondria and ribosomes.

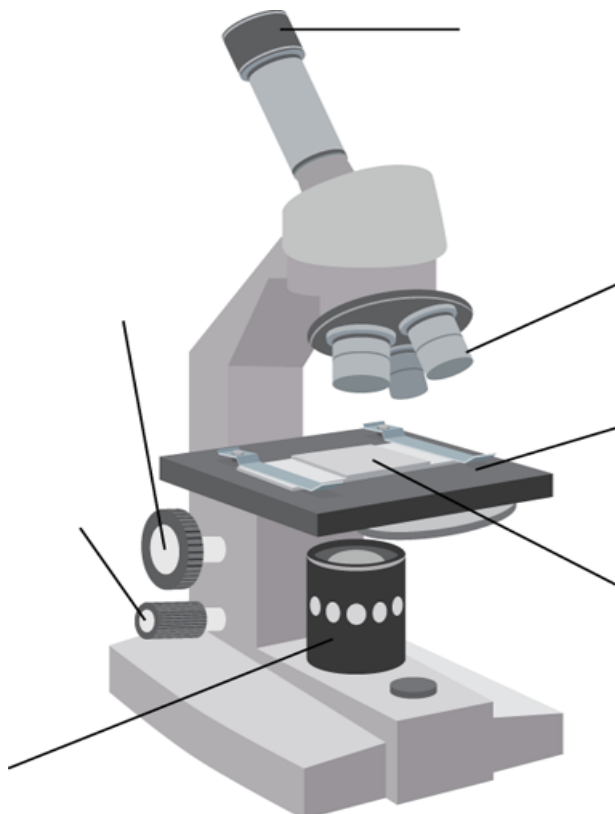
Which organelles can be found in only plant cells?

Chloroplasts, a cellulose cell wall and a large permanent vacuole.

## Parts of a Light Microscope

Label the microscope using the following words:

**coarse focus,      fine focus,      stage,      light,      slide,      objective lens,      eyepiece**



Explain the functions of the following parts of the microscope:

**Lamp:** Provides the light that shines up through the specimen so that it can be seen.....

**Focusing knob:** Moves the stage (or the lens) up and down to bring the specimen into focus -- coarse/rough and fine.

**Stage:** The platform that the slide with the specimen is placed on.....

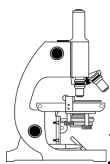
**Objective lens:** The lens closest to the specimen; it magnifies the specimen (e.g.  $\times 4$ ,  $\times 10$ ,  $\times 40$ ).....

Which objective lens would you use to find your specimen?

The low power objective lens ( $\times 4$ ).....

Which objective lens would you use to observe detail?

The high power objective lens ( $\times 40$ ).....



## **Using a Microscope**

*(Never touch the surfaces of the eyepiece or objective lens, use specialized lens tissue if they are dirty)*

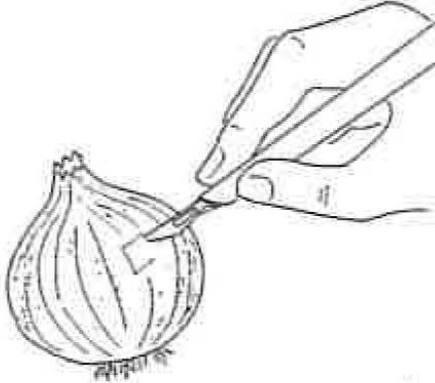
**Number the following statements in the correct order:**

| Order | Instructions                                                                                                                                                                |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9     | Once your specimen is in focus, you can increase the magnification by turning the nosepiece so that a higher power objective lens is above the slide                        |
| 3     | Use the rough focusing knob to move the stage to its lowest position                                                                                                        |
| 4     | Place your slide on to the stage                                                                                                                                            |
| 6     | Look through the eyepiece lens                                                                                                                                              |
| 7     | Locate your specimen using the rough focusing knob (you may need to adjust the position of the slide by moving the stage)                                                   |
| 8     | Adjust the focus for the enlarged image by using the fine focusing knob – do NOT use the rough focusing knob at this point as you could damage the lens or crack the slide! |
| 2     | Turn the nosepiece so that the low power objective lens clicks into position above the hole in the stage                                                                    |
| 5     | Secure your slide with the stage clips                                                                                                                                      |
| 1     | Switch on your microscope                                                                                                                                                   |

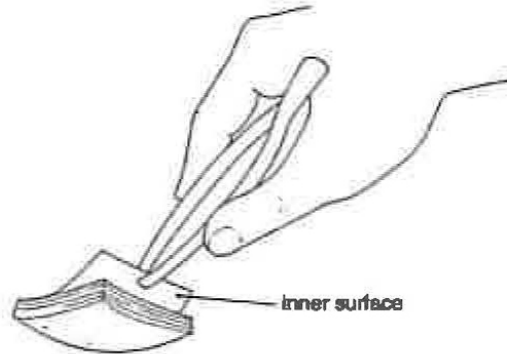
**Once you have the steps in the correct order, you are ready to begin looking at real slides!**

# Looking at plant cells

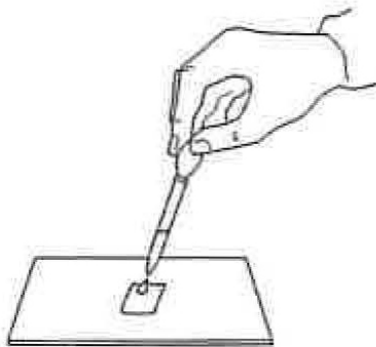
Before you look at onion cells under the microscope, you must peel off a very thin layer.

**Step 1**

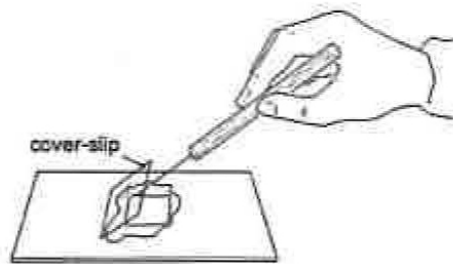
- **Carefully** cut out a small piece of onion tissue.

**Step 2**

- Use forceps or your finger nails to peel off the inner surface (this looks like tissue paper).

**Step 3**

- Put the piece of onion 'skin' flat on a slide and add two drops of iodine solution.

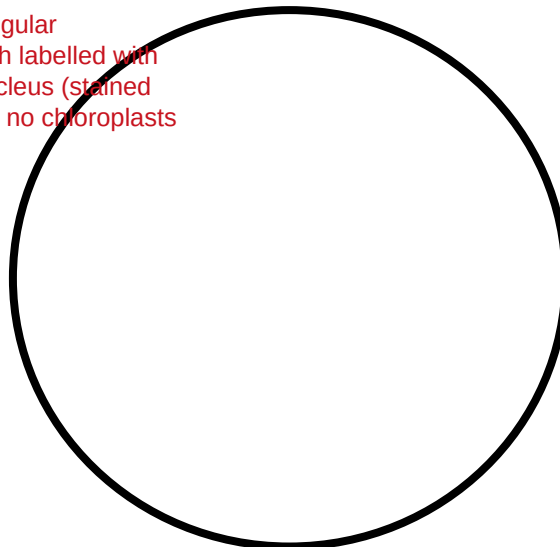
**Step 4**

- Gently lower the cover-slip onto the slide using a mounted needle.
- Now look at your onion cells under the microscope.

**Draw what your onion cells look like here:**

Don't forget to label any organelles.

• Onion epidermis cells drawn as rectangular 'brick-shaped' cells joined together, each labelled with cell wall, cell membrane, cytoplasm, nucleus (stained darker) and a large permanent vacuole; no chloroplasts (onion is not green).



## Looking at animal cells

To avoid any chance of infection carry out these instructions carefully.



### Step 1



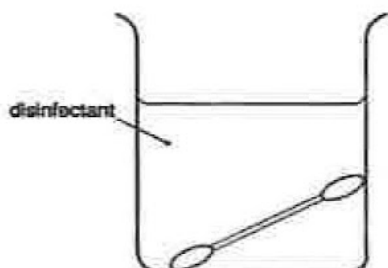
- Your teacher will give you a cotton bud from a freshly opened pack.
- Gently wipe the lining of your cheek with one end of the cotton bud.

### Step 2



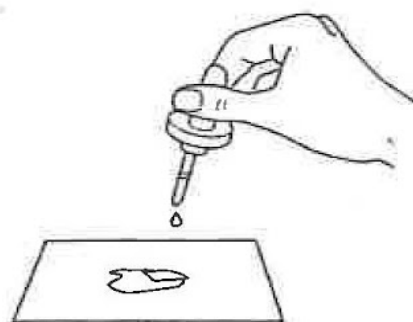
- Smear the cotton bud over the centre of a slide.

### Step 3



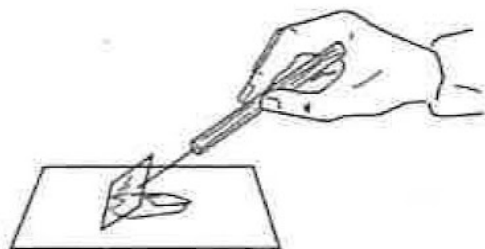
- Immediately dispose of the cotton bud into a beaker of disinfectant provided by your teacher.

### Step 4



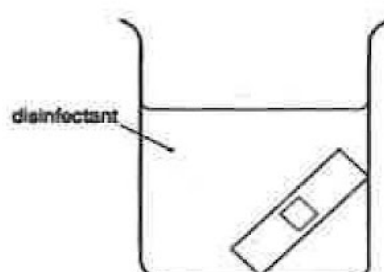
- Put a drop of methylene blue stain on top of the smear.

### Step 5



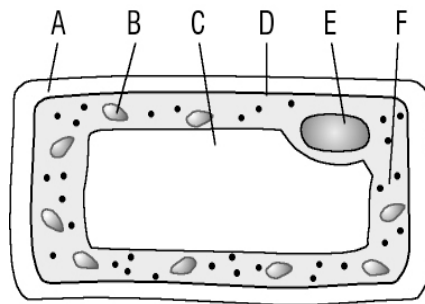
- Place a cover-slip on top and look for cells under the microscope at high power.
- Draw some of your cheek cells (2 or 3) and label your drawing.

### Step 6



- At the end of the activity put the slide with its cover-slip into a beaker of disinfectant.

**Q1** The diagram is of a cell from the leaf of a plant.



(a) Name the structures **D**, **E** and **F**.

nucleus.....

cell membrane.....

cytoplasm.....

(3)

(b) (i) What is the name of structure **A**?

cell wall.....(1)

(ii) What material is structure **A** made of?

cellulose.....(1)

(c) (i) What is the name of structure **C**?

vacuole.....(1)

(ii) What is the liquid it contains called?

cell sap.....(1)

(d) Structure **B** is a chloroplast. What is its function?

Absorbs light energy needed for photosynthesis.....

(it contains chlorophyll which traps the light energy used to make food).....

(2)

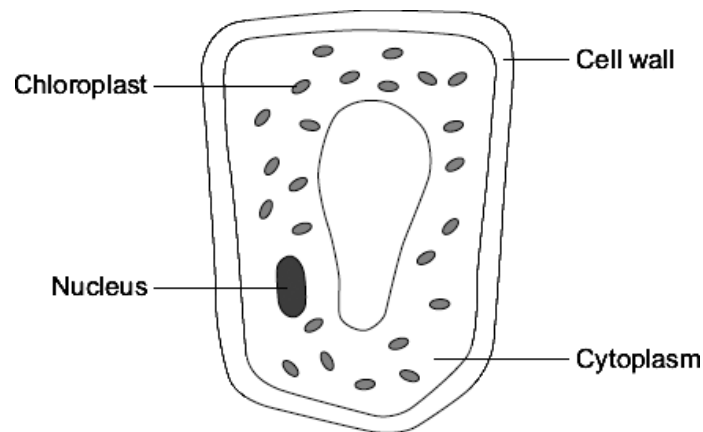
(e) Name two different structures that are found within the material labelled **F**.

mitochondria.....

ribosomes.....

(2)

**Q2.** The diagram shows a plant cell from a leaf.



(a) **List A** gives the names of three parts of the cell.

**List B** gives the functions of parts of the cell.

Draw a line from each part of the cell in **List A** to its function in **List B**.

**List A**  
**Parts of the cell**

Draw lines: Nucleus -> Controls the activities of the cell; Cytoplasm -> Where most of the chemical

reactions take place; Chloroplast -> Absorbs light energy to make food. (Strengthens the cell is the

Nucleus

Cytoplasm

Chloroplast

**List B**  
**Functions**

Where most of the chemical reactions take place

Absorbs light energy to make food

Strengthens the cell

Controls the activities of the cell

(3)

(b) Respiration takes place in the cell.

Draw a ring around the correct answer to complete the sentence.

All cells use respiration to release

energy

oxygen.

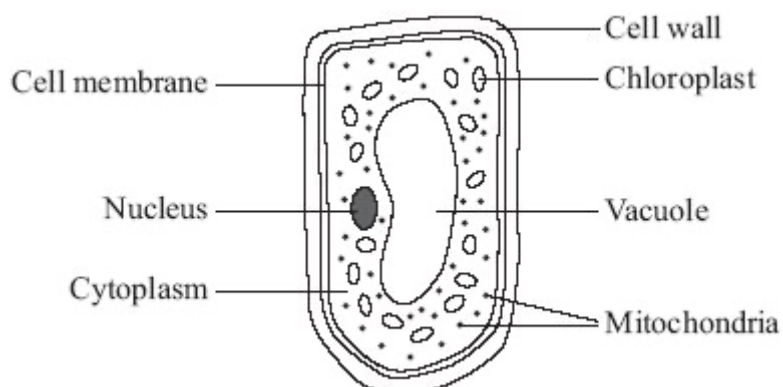
sugar.

(1)

(Total 4 marks)

**Q3. Diagram 1** shows a cell from a leaf.

**Diagram 1**



(a) How is the leaf cell specialised to carry out photosynthesis?

Tick (✓) **one** box.

It has a permanent vacuole.

☐

It has many chloroplasts.

☒

It has cytoplasm.

☐

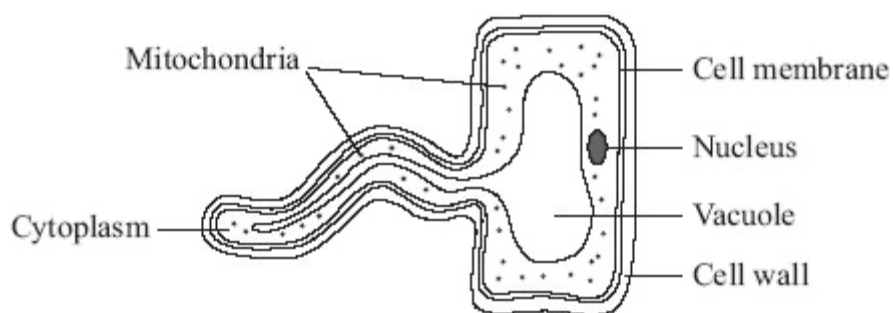
It has many mitochondria.

☐

(1)

(b) **Diagram 2** shows another type of plant cell.

**Diagram 2**



Give **two** ways in which this cell is different from an animal cell.

1. 1. It has a cell wall made of cellulose

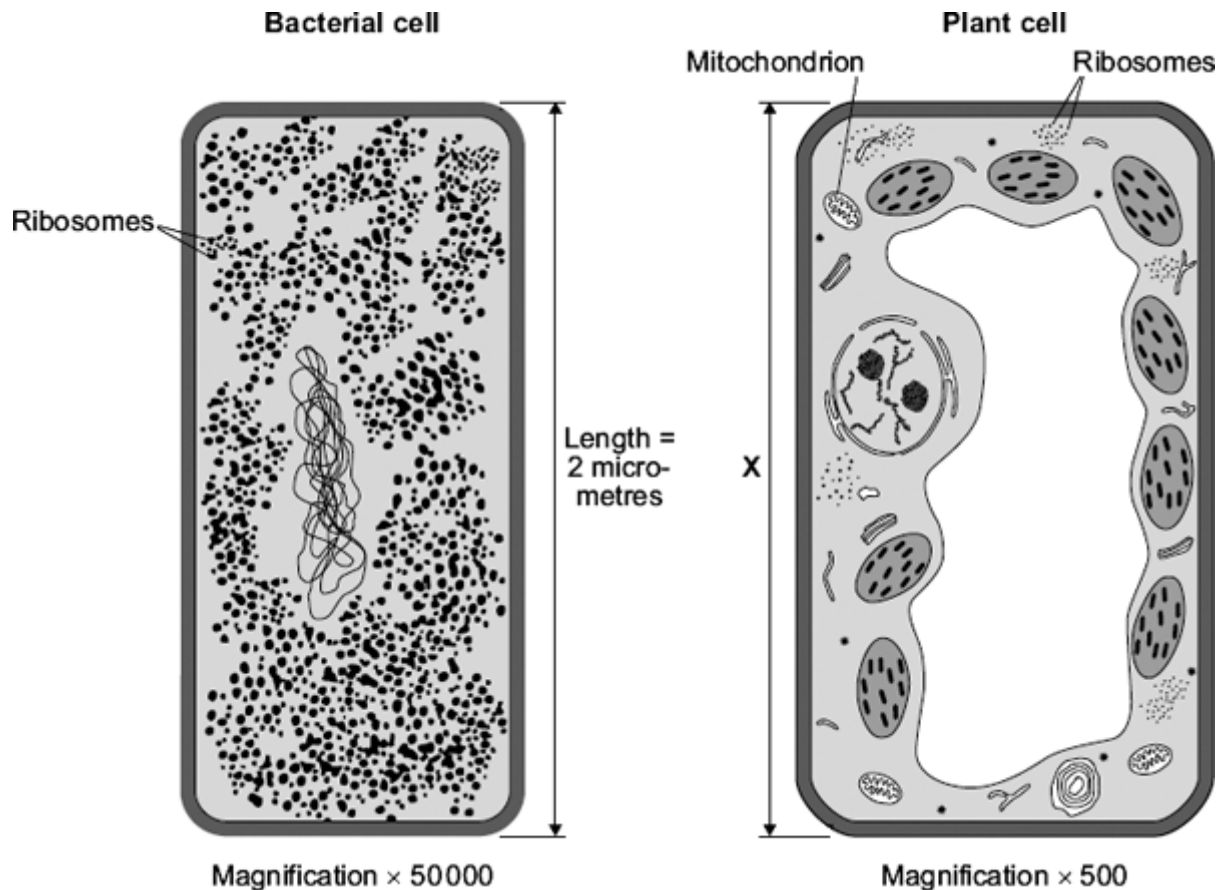
which gives the cell a fixed shape and support.

2. 2. It has a large permanent vacuole

filled with cell sap, which helps keep the cell firm (turgid).

(2)

**Q4.** The diagram shows two cells, a bacterial cell and a plant cell.



- (a) (i) Both the bacterial cell and the plant cell contain ribosomes.

What is the function of a ribosome?

It is where protein synthesis takes place - proteins are made.

Ribosomes are found in the cytoplasm of the cell.

(1)

- (ii) The plant cell contains mitochondria but the bacterial cell does **not** contain mitochondria.

Give **one** other way in which the plant cell is different from the bacterial cell.

The plant cell has a nucleus

but the bacterial cell has no nucleus (its DNA is loose in the cytoplasm).

(1)

(Total 2 marks)

### Lesson 3: Diffusion

What substances does a cell need?

1. What useful substances need to move into a cell?

~~Oxygen and glucose (also mineral ions, amino acids and water)~~.....

2. What waste products does a cell need to get rid of?

~~Carbon dioxide (and urea)~~.....

What is the definition of diffusion?

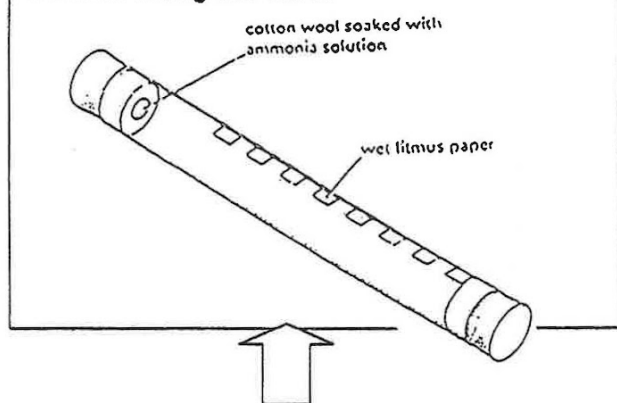
~~Diffusion is the net movement of particles from an area of higher concentration~~.....

~~to an area of lower concentration (down a concentration gradient)~~.....

~~It is a passive process -- no energy is needed and it happens because of random movement~~.....

## Proving diffusion!

This apparatus will prove diffusion happens. The wet litmus paper will turn blue as the ammonia solution turns into gas and diffuses along the tube.



1. Q. Can you draw an arrow to show the Direction of diffusion.

Q, Can you label each end of the tube with high concentration and low concentration.

2. So what if we made the concentration of the ammonia solution stronger (more particles!). What do you think will happen to the diffusion speed. (CIRCLE the answer you think in red.)

Speed up

Slow down

Same

3. Now record the time for all the litmus strips to turn blue for the weak ammonia and strong ammonia.

Weak (time) \_\_\_\_\_ Strong (time) \_\_\_\_\_

4. Well! Were you right? If NOT circle the correct choice in blue.

Now...can you explain why below?

Concentration affects diffusion speed because.....

**Effect of size on uptake by diffusion****SAFETY:**

Take care with the solutions used: wear eye protection and rinse splashes off the skin.

**PROCEDURE:**

- Collect agar cubes of different sizes – 2 cm x 2 cm x 2 cm and 1 cm x 1 cm x 1 cm
- Place the cubes in a beaker and cover with the acid (just enough to cover the surface).
- Start the stop clock.
- Leave the cubes and measure the time taken for the cubes to lose their pink colour completely.
- While you are waiting, complete the table with calculations for each cube. (See student sheet.)

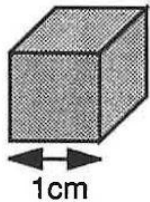
| A<br>Length<br>of side<br>of agar<br>cube<br>(cm)                                                                                                         | B<br>= A <sup>2</sup><br>Area of<br>one side<br>of cube<br>(cm <sup>2</sup> ) | C<br>= 6B<br>Total<br>surface<br>area of<br>cube<br>(cm <sup>2</sup> ) | D<br>= A <sup>3</sup><br>Volume of<br>cube<br>(cm <sup>3</sup> ) | E<br>= C÷D<br>Surface<br>area to<br>volume<br>ratio | F<br>Distance<br>travelled by<br>acid (clear)<br>in mm |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| 1                                                                                                                                                         |                                                                               |                                                                        |                                                                  |                                                     |                                                        |
| Row 1 (1 cm cube): A = 1, B = 1 cm <sup>2</sup> , C = 6 cm <sup>2</sup> , D = 1 cm <sup>3</sup> , E = 6, F = measured from the experiment (about 5 mm).   |                                                                               |                                                                        |                                                                  |                                                     |                                                        |
| 2                                                                                                                                                         |                                                                               |                                                                        |                                                                  |                                                     |                                                        |
| Row 2 (2 cm cube): A = 2, B = 4 cm <sup>2</sup> , C = 24 cm <sup>2</sup> , D = 8 cm <sup>3</sup> , E = 3, F = measured from the experiment (about 5 mm).  |                                                                               |                                                                        |                                                                  |                                                     |                                                        |
| 3                                                                                                                                                         |                                                                               |                                                                        |                                                                  |                                                     |                                                        |
| Row 3 (3 cm cube): A = 3, B = 9 cm <sup>2</sup> , C = 54 cm <sup>2</sup> , D = 27 cm <sup>3</sup> , E = 2, F = measured from the experiment (about 5 mm). |                                                                               |                                                                        |                                                                  |                                                     |                                                        |

How to calculate surface area and volume:

**Cube A**

SA of face = 1 cm x 1 cm = 1 cm<sup>2</sup>  
 SA of cube = 1 cm<sup>2</sup> x 6 = 6 cm<sup>2</sup>  
 V of cube = 1 cm x 1 cm x 1 cm = 1 cm<sup>3</sup>  
 SA/V = 6:1

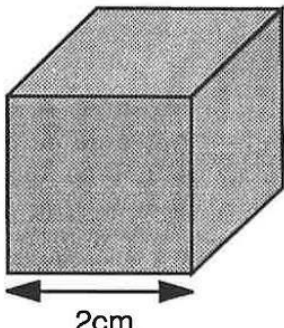
A



**Cube B**

SA of face = 2 cm x 2 cm = 4 cm<sup>2</sup>  
 SA of cube = 4 cm<sup>2</sup> x 6 = 24 cm<sup>2</sup>  
 V of cube = 2 cm x 2 cm x 2 cm = 8 cm<sup>3</sup>  
 SA/V = 3:1

B



Volume of a cube = length x width x height (side cubed). Surface area of a cube = 6 x (side x side), because a cube has 6 faces.

# QUESTIONS:

- 1 Compare the surface area of the 1 cm block to the 2 cm block. In each case the length of the side of the block doubles. What happens to the surface area?

~~The surface area is 4 times bigger (it increases by a factor of 4): 1 cm cube = 6 cm<sup>2</sup>, 2 cm cube = 24 cm<sup>2</sup>.....~~

- 2 Compare the volume of the 1 cm block to the 2 cm block. In each case the length of the side of the block doubles. What happens to the volume?

~~The volume is 8 times bigger (it increases by a factor of 8): 1 cm<sup>3</sup> becomes 8 cm<sup>3</sup>.....~~

- 3 Compare the surface area to volume ratio of the 1 cm block to the 2 cm block. In each case the length of the side of the block doubles. What happens to the surface area to volume ratio?

~~The surface area to volume ratio decreases by half... it falls from 6:1 to 3:1, because the volume grows faster than the surface area.~~

- 4 Is there any difference in the rate of diffusion as the size of the agar cubes increases?

~~Yes. The acid takes longer to reach the centre of the larger cube, because the distance is greater and the surface area to volume ratio is smaller.~~

- 5 The volume of a living organism is proportional to the number of cells in its body. Each cell needs oxygen and nutrients and needs to get rid of metabolic waste products such as carbon dioxide. The smallest organisms can absorb nutrients and get rid of waste by diffusion through their outer membranes. Do you think larger organisms can do the same? Why?

~~No. Larger organisms have a small surface area to volume ratio and the distance to the centre of the body is too large, so...~~

~~Diffusion would need a transport system (such as blood) to carry substances quickly to all the cells.....~~

- 6 Should this procedure have a control? If so, what would it be?

~~Yes. A cube of the same size in water instead of acid, to show that the acid is what causes the colour to be lost.~~

~~This is a control because everything else (size of cube, time, temperature) is kept the same.....~~

- 7 Complete the following paragraph, using the words provided below. Each word may be used once, more than once, or not at all.

|                 |                 |                      |                     |             |                  |
|-----------------|-----------------|----------------------|---------------------|-------------|------------------|
| <b>Third</b>    | <b>decrease</b> | <b>body surfaces</b> | <b>second</b>       | <b>area</b> | <b>diffusion</b> |
| <b>relative</b> | <b>ratio</b>    | <b>volume</b>        | <b>surface area</b> | <b>slow</b> | <b>size</b>      |

Living organisms exchange materials across body surfaces by diffusion. Increasing size

means that volume grows faster than surface area. Exchanging materials by diffusion

becomes a very slow process and not enough material will reach cells buried deep within

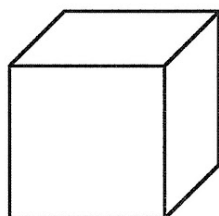
the body of large organisms. Complex structures like the root system of a plant or the lungs

of a mammal provide a large surface area for exchange and short distances for diffusion.

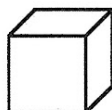
- 8 A student investigated the effect of size on the movement of molecules.

He cut three different sized cubes from a block of clear agar jelly.

Cube A was  $2 \times 2 \times 2$  cm. Cube B was  $1 \times 1 \times 1$  cm and cube C was  $0.5 \times 0.5 \times 0.5$  cm.



cube A



cube B



cube C

The student wore safety glasses and placed the cubes in a beaker of red dye solution for five minutes.

He then poured the solution into another beaker, washed the surface of the cubes and dried them with blotting paper.

He then cut each cube in half and examined the newly cut surfaces.

Diagram 1 shows what the cubes looked like.

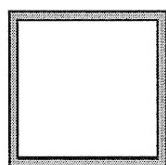


Diagram 1

- a. Why did he wear safety glasses? (1)

~~The dye/acid solution is an irritant, so safety glasses protect his eyes from splashes.~~.....

~~This reduces the risk of the solution damaging his eyes.~~.....

- b. Explain how the red dye molecules entered the jelly. (2)

~~The dye particles diffuse into the jelly from an area of higher concentration to an area of lower concentration.~~.....

~~The particles move randomly down the concentration gradient -- no energy is needed.~~.....

- c. Use a ruler to measure the distance the red dye has entered each cube in diagram 1. (1)

Answer: ~~5~~.....mm

- d. Calculate the surface area of cube A. (2)

Surface Area= 6.....unit cm<sup>2</sup>.....

- (ii) Calculate the volume of cube A.

volume= 1..... unit cm<sup>3</sup>.....

- e. The student was told by his teacher that the cubes have different surface area to volume ratios.

Complete the table by putting one tick (✓) in each row to show whether the statement applies to cube A, B or C. (3)

|                                          | ✓ | Cube A | Cube B | ✓ | Cube C |
|------------------------------------------|---|--------|--------|---|--------|
| Largest surface area                     |   |        |        |   |        |
| Largest surface area to volume ratio     |   |        |        |   |        |
| Greatest proportion of cube coloured red |   |        |        |   |        |

- f. Explain how this experiment can be used to understand the need to transport systems in larger organisms such as humans. (3)

~~Large organisms have a small surface area to volume ratio, so the distance substances must diffuse from the surface to the centre of the body is too great. In the experiment, the red dye only reached the outside of the biggest cube, showing diffusion is too slow to supply the middle. Diffusion alone would not deliver enough oxygen and nutrients to the cells deep inside the body, and wastes would build up. So larger organisms need a transport system (heart, blood, capillaries) to move substances quickly to every cell.~~

(Total for Question 8=14 marks)

## Lesson 4 Osmosis

Define these terms:

Dilute

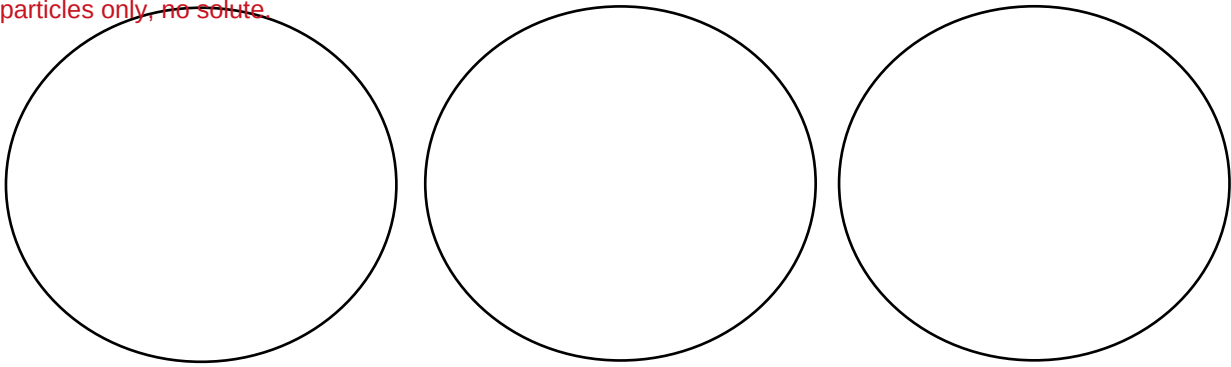
~~A dilute solution contains a lot of water and only a little dissolved solute (e.g. sugar).~~.....

Concentrated

~~A concentrated solution contains a lot of dissolved solute and less water.~~.....

Draw a dilute solution, a concentrated solution and a pure water diagram in each of the spaces below.

~~Draw particles in the boxes: dilute = a few solute dots in lots of water; concentrated = many solute dots; pure water = water particles only, no solute.~~



Define osmosis

~~Osmosis is the movement of water molecules through a partially permeable membrane, from a dilute solution (high water concentration) to a concentrated solution (low water concentration). It is a passive process - no energy is needed.~~.....

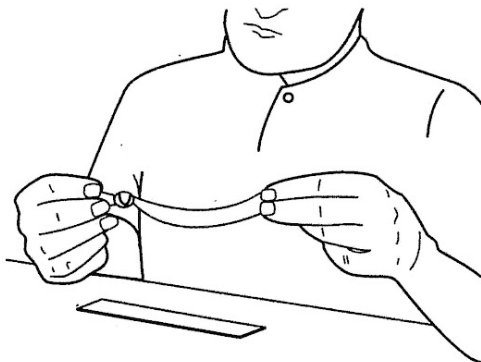
What is a partially permeable membrane?

~~A membrane with very small holes that lets small molecules such as water pass through, but not larger solute molecules.~~.....

***A model cell***

Copy and complete this table:

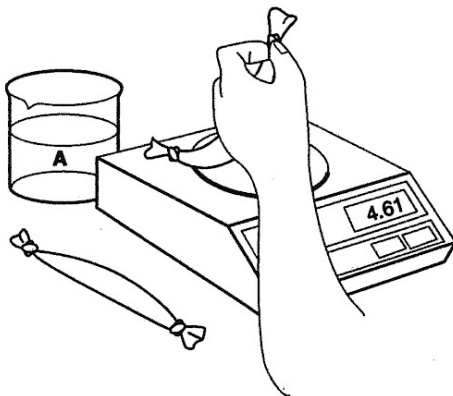
| model cell | liquid inside<br>Visking tubing | liquid in<br>beaker | weight of model cell (g) |               | change in<br>weight of cell (g) |
|------------|---------------------------------|---------------------|--------------------------|---------------|---------------------------------|
|            |                                 |                     | at start                 | after 30 mins |                                 |
| A          |                                 |                     |                          |               |                                 |
| B          |                                 |                     |                          |               |                                 |

**Step 1**

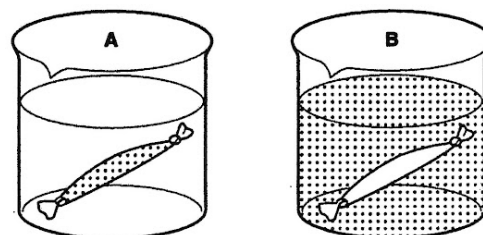
- Cut 2 pieces of Visking tubing 12 cm long. Tie the end of each with cotton.

**Step 2**

- Fill cell A with sugar solution and tie the open end with cotton.
- Fill cell B with water and tie the open end with cotton.

**Step 3**

- Wipe the outside of each model cell carefully with a paper towel.
- Weigh each cell and put the reading in the table.

**Step 4**

- Put cell A into a beaker of water and cell B into a beaker of sugar solution.
- Re-weigh your model cells after 30 minutes.

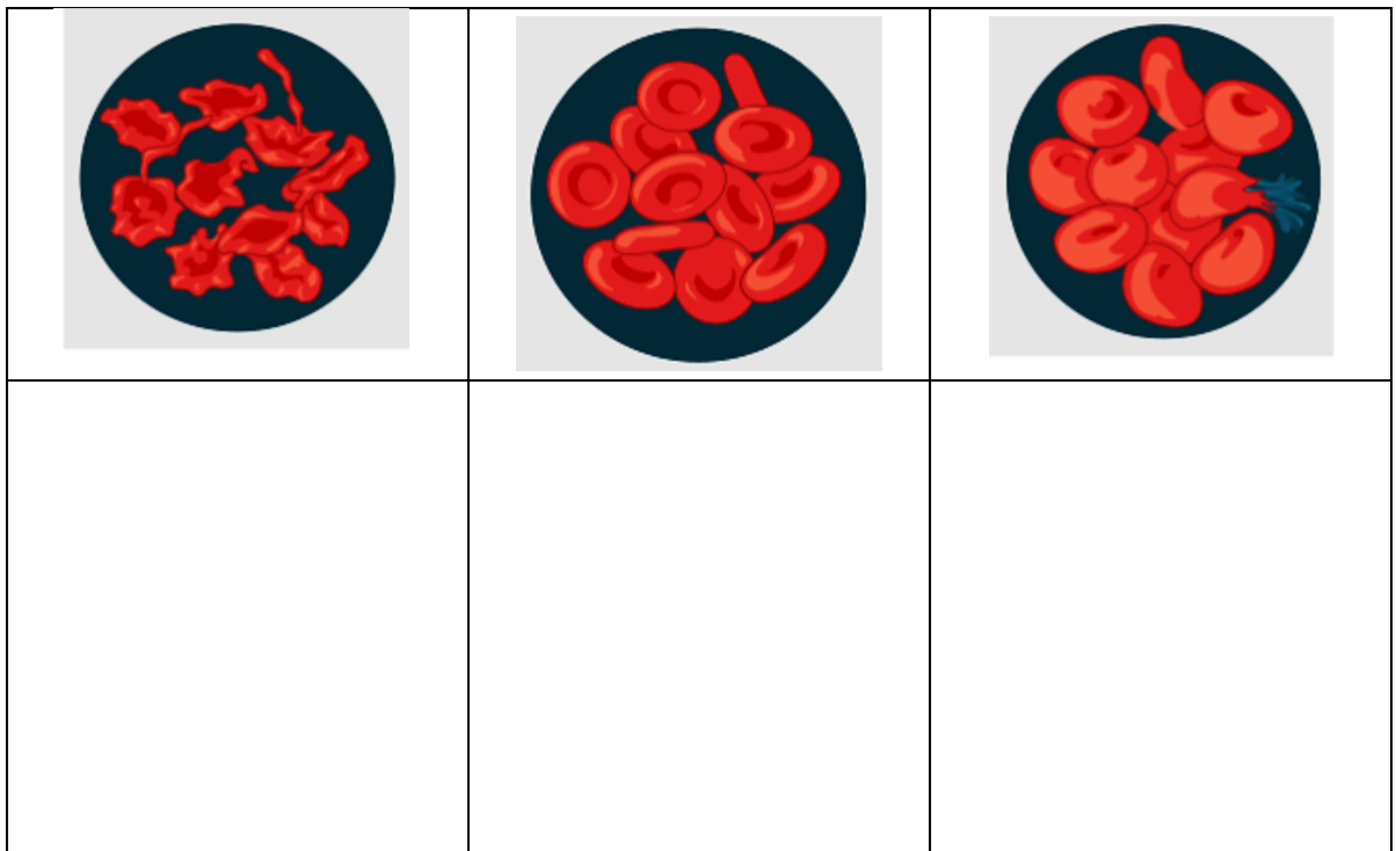
**Questions****1** What did cell A look like after 30 minutes?**2** What has happened to the weight of cell A after 30 minutes?

Explain what has happened.

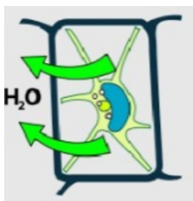
**3** What did cell B look like after 30 minutes?**4** What has happened to the weight of cell B after 30 minutes?

Explain what has happened.

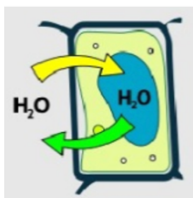
Fill in the cartoon strip to show describe what happens when you place an animal cell into an isotonic (same concentration), a concentrated and a dilute solution.



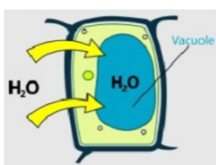
Match the diagram to the correct description of how a plant cell behaves in each type of solution:



Plant cell placed in a concentrated solution. This leads to the movement of water out of the cell. This can lead to plasmolysis.



Plant cell placed in a dilute solution. This leads to the movement of water into the cell causing turgor pressure.



Plant cell placed in a isotonic solution. Water enters and leaves the plant cell at an equal rate, this leaves the plant cell flaccid.

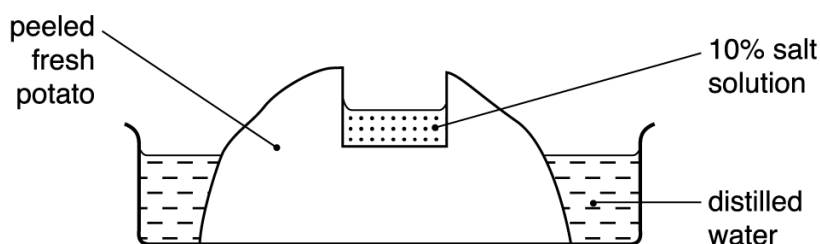
· Isotonic - the cell stays the same size (water moves in and out at equal rates). Concentrated - water leaves the cell by osmosis, so it shrinks and shrivels. Dilute - water enters the cell by osmosis, so it swells and may burst.

· Match the diagrams: cell with the membrane pulled away from the wall (plasmolysed) = concentrated solution; firm, swollen cell (turgid) = dilute solution; cell that is soft/flaccid with no change = isotonic solution.

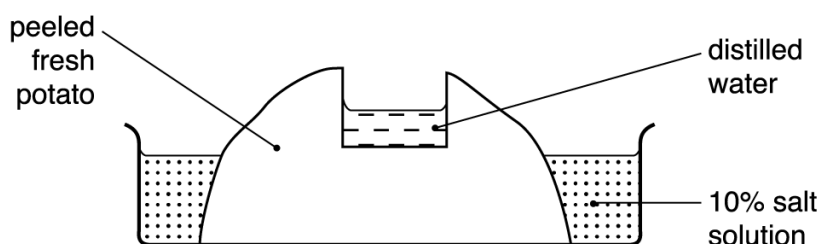
## Potato Osmometer

The apparatus in the diagram is used to demonstrate the movement of water through plant tissue.

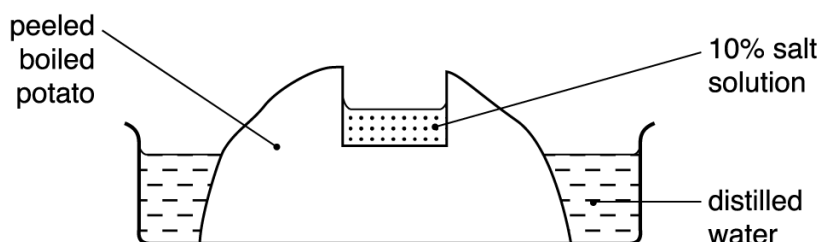
### apparatus A



### apparatus B



### apparatus C



Describe what changes you would expect to observe after leaving the apparatus set up for one hour, and, in each case, explain why the change has occurred.

#### Apparatus A

Change ~~The water level in the capillary tube rises (the potato gains mass).~~.....

Explanation ~~Water moves into the potato cells by osmosis, because the water concentration outside is higher than inside the cells.~~

#### Apparatus B

Change. ~~The water level stays about the same +/- changes very little.~~.....

Explanation ~~The solution and the potato cells are at about the same concentration (isotonic); so water moves in and out at equal rates.~~

#### Apparatus C

Change ~~The water level in the capillary tube falls (the potato loses mass).~~.....

Explanation ~~Water moves out of the potato cells by osmosis into the concentrated sugar solution, which has a lower water concentration.~~

## Lesson 5: Investigate the effect of sugar solution on plant tissue

In this practical you should:

- Identify independent, dependent and control variables
- Write a risk assessment
- Produce a graph of your results
- Write a conclusion using your results and your scientific knowledge

### Equipment

- Potatoes
- Scalpel
- Ruler
- 5 boiling tubes
- Sugar solution – 0 mol dm<sup>-3</sup>, 0.2 mol dm<sup>-3</sup>, 0.4 mol dm<sup>-3</sup>, 0.6 mol dm<sup>-3</sup>, 0.8 mol dm<sup>-3</sup>, 1 mol dm<sup>-3</sup>
- Measuring cylinder
- Sticky labels
- Test tube rack
- Kitchen roll
- Balance

### Method

1. Cut the potato into six equally sized cubes (using the ruler) and pat the cubes dry.
2. Use the balance to find the start mass of the cubes.
3. Put the boiling tubes into the rack and put a cube of potato in each tube.
4. Use the measuring cylinder to measure 20cm<sup>3</sup> of the first solution into one of the tubes, label the tube to show which solution is in it.
5. Repeat with the other solutions.
6. Leave for the amount of time suggested by your teacher.
7. Remove the cubes from the boiling tubes and re-weigh them.
8. Calculate the change in mass for each cube.

### Risk assessment

1. What is the main risk in this experiment?

~~The main risk is cutting yourself with the scalpel while cutting the potato.~~

2. How could you reduce this risk?

~~Cut on a white tile, keep the blade pointing away from your fingers, and keep your fingers behind the blade.~~

### Variables

3. What is the independent variable in this experiment?

~~The concentration of the sugar solution (0.0 to 1.0 mol dm<sup>-3</sup>).~~

4. What is the dependent variable in this experiment?

~~The change in mass of the potato cube (end mass – start mass, or % change in mass).~~

5. What is a control variable in this experiment?

~~The size (volume) of the potato cubes – also the time left and the temperature.~~

6. How is this variable kept the same?

~~Cut all six cubes to the same size using a ruler, and dry them the same way.~~

~~and leave them in the solutions for the same length of time at the same temperature.~~

**Results**

| Concentration (mol dm <sup>-3</sup> ) | Mass of potato at start (g) | Mass of potato at end (g) | Change in mass (g) |
|---------------------------------------|-----------------------------|---------------------------|--------------------|
| 0.0                                   | 2.50                        | 2.85                      | +0.35              |
| 0.2                                   | 2.50                        | 2.72                      | +0.22              |
| 0.4                                   | 2.50                        | 2.50                      | 0.00               |
| 0.6                                   | 2.50                        | 2.35                      | -0.15              |
| 0.8                                   | 2.50                        | 2.20                      | -0.30              |
| 1.0                                   | 2.50                        | 2.05                      | -0.45              |

**Draw a graph of your results on the next page**

**Analysis**

1. Which variable goes on your x-axis?

~~The concentration of the sugar solution (mol dm<sup>-3</sup>)~~.....

2. Which variable goes on your y-axis?

~~The change in mass of the potato cubes (g or % change)~~.....

3. Why do you calculate % change?

~~Because the cubes may not have started with exactly the same mass, so the percentage change lets you compare them fairly.~~

4. In the diagram shown, what is the arrow showing moving out of the potato?

~~Water~~.....

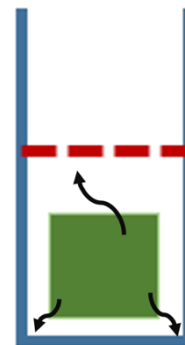
5. What is this process known as?

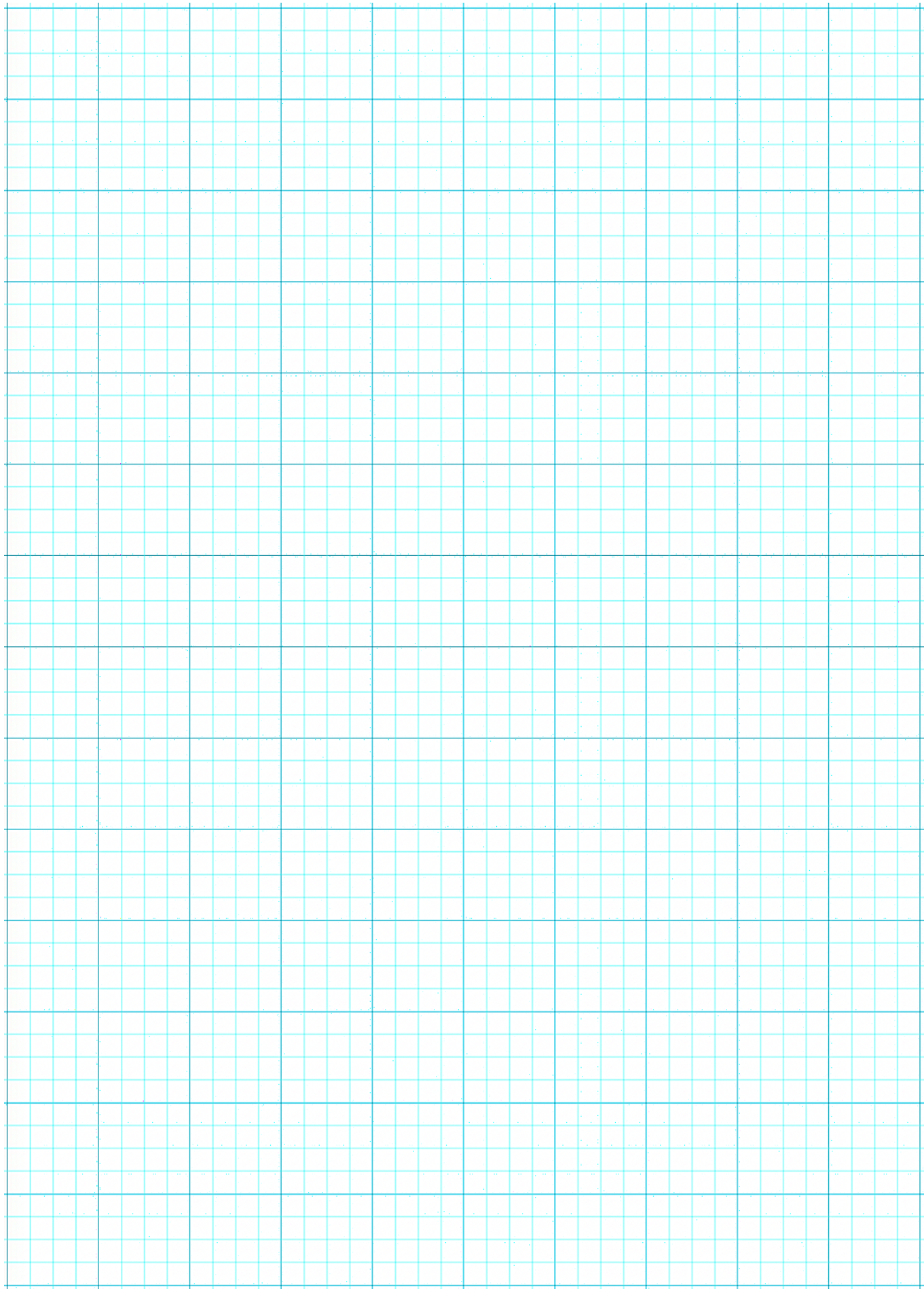
~~Osmosis~~.....

6. Write out the definition of the process

~~Osmosis is the movement of water molecules from a dilute solution (high water~~.....

~~concentration) to a concentrated solution (low water concentration) through a partially permeable membrane.~~





· Graph page for the potato practical: plot sugar solution concentration against change in mass of the potato from the results.

## Conclusion

1. Describe the relationship between sugar solution and mass of the potato shown in your graph.

~~As the sugar concentration increases, the change in mass of the potato decreases; in dilute solution the potato...~~  
~~gains mass, in concentrated solution it loses mass.~~.....  
.....  
.....

2. Explain the relationship between sugar solution and mass of the potato.

~~Water moves from high to low water concentration through the partially permeable membrane; concentrated~~.....  
~~sugar solution draws water out (mass falls), dilute solution lets water in (mass rises).~~.....  
.....  
.....

## Lesson 6: Active Transport

Define active transport:

Active transport is the movement of substances against a concentration gradient, from a region of lower concentration to a region of higher concentration, using energy released by respiration.

Watch the video on active transport and answer the following questions:

<https://www.youtube.com/watch?v=eDeCgTRFCbA>

1. What is required in order for active transport to take place?

Energy (released by respiration).

2. What is the name of the structure which carries molecules across the cell membrane during active transport?

Carrier proteins in the cell membrane.

3. Which process released the energy needed for active transport?

Respiration (aerobic respiration).

4. Give an example of a plant cell where active transport commonly takes place.

Root hair cells (absorbing mineral ions from the soil).

5. Which organelle is prevalent (more common) in cells that carry out active transport more frequently?

Mitochondria.

## Comparing diffusion, osmosis and active transport.

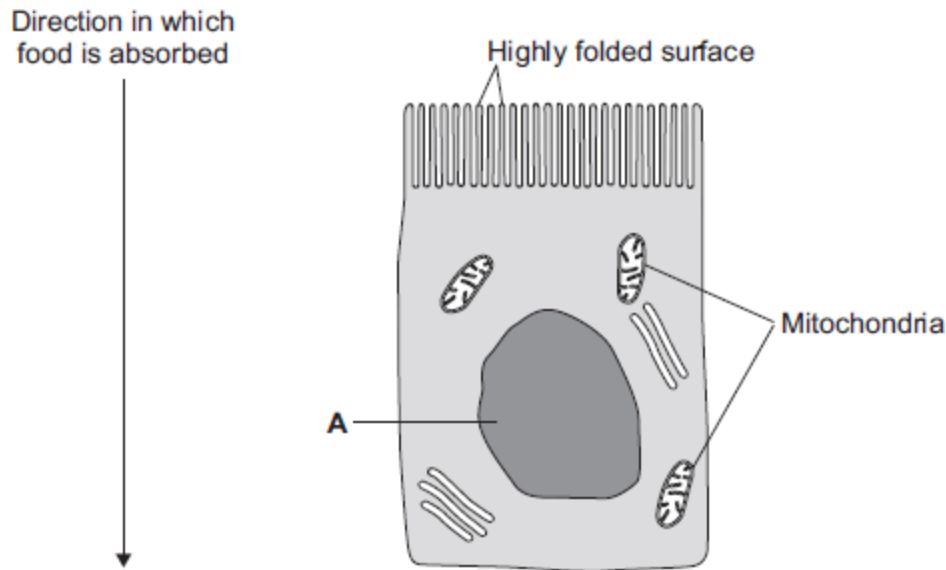
**Task:** Sort the following statements into the correct columns.

- Passive process
- Active process
- Requires energy
- Does not require energy
- Moves molecules against a concentration gradient
- Move molecules down a concentration gradient
- Moves substances from a high to a low concentration
- Moves substances from a low to a high concentration
- Movement of water only

| Diffusion                                                                                                                              | Osmosis                                                                                                                                     | Active transport                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Active process; Requires energy; Moves molecules against a concentration gradient; Moves substances from a low to a high concentration | Passive process; Does not require energy; Move molecules down a concentration gradient; Moves substances from a high to a low concentration | Passive process; Does not require energy; Move molecules down a concentration gradient; Moves substances from a high to a low concentration; Movement of water only |

### Exam Practice Questions

The image below shows an epithelial cell from the lining of the small intestine.



- (a) (i) In the image above, the part of the cell labelled **A** contains chromosomes.

What is the name of part **A**?

Nucleus

(1)

- (ii) How are most soluble food molecules absorbed into the epithelial cells of the small intestine?

Draw a ring around the correct answer.

|                  |         |             |
|------------------|---------|-------------|
| <u>diffusion</u> | osmosis | respiration |
|------------------|---------|-------------|

(1)

- (b) Suggest how the highly folded cell surface helps the epithelial cell to absorb soluble food.

It increases the surface area of the cell membrane, so more soluble food molecules can be absorbed more quickly (by diffusion/active transport).

(1)

- (c) Epithelial cells also carry out active transport.

- (i) Name **one** food molecule absorbed into epithelial cells by active transport.

Glucose

(1)

- (ii) Why is it necessary to absorb some food molecules by active transport?

The concentration of these molecules inside the cells is higher than in the gut, so they must be moved against the concentration gradient using energy from respiration.

(1)

- (ii) Suggest why epithelial cells have many mitochondria.

Active transport needs energy, and mitochondria release energy by respiration, so many mitochondria provide enough energy for the cell to absorb food molecules.

---



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(2)

- (d) Some plants also carry out active transport.

Give **one** substance that plants absorb by active transport.

Mineral ions (e.g. nitrate ions), absorbed from the soil into root hair cells.

(1)

Chicken eggshells contain calcium. Calcium ions are moved from the shell into the cytoplasm of the egg.

**Table 2** shows information about the concentration of calcium ions.

**Table 2**

| Location      | Concentration of calcium ions in arbitrary units |
|---------------|--------------------------------------------------|
| Egg shell     | 0.6                                              |
| Egg cytoplasm | 2.1                                              |

- (e) Explain how calcium ions are moved from the shell into the cytoplasm of the egg.

Calcium ions are moved by active transport: the cytoplasm (2.1) has a higher concentration than the shell (0.6), so the ions move against the concentration gradient, using energy.

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(3)

- Q2.** (a) Some scientists investigated the rates of absorption of different sugars by the small intestine.

In one experiment they used a piece of normal intestine.

In a second experiment they used a piece of intestine poisoned by cyanide. Cyanide is poisonous because it prevents respiration.

The results are shown in the table.

| Sugar     | Relative rates of absorption |                               |
|-----------|------------------------------|-------------------------------|
|           | Normal intestine             | Intestine poisoned by cyanide |
| Glucose   | 1.00                         | 0.33                          |
| Galactose | 1.10                         | 0.53                          |
| Xylose    | 0.30                         | 0.31                          |
| Arabinose | 0.29                         | 0.29                          |

- (i) Name **two** sugars from the table which can be absorbed by active transport.

1. Glucose

2. Galactose

(1)

- (ii) Use evidence from the table to explain why you chose these sugars.

Cyanide stops respiration, so less energy for active transport: glucose (1.00 to 0.33) and galactose (1.10 to 0.53) are absorbed much more slowly, while xylose and arabinose barely change.

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(3)

- (b) All of the sugars named in the table can be absorbed by diffusion. Explain how information from the table provides evidence for this.

All four sugars are still absorbed (their rates are greater than zero) even when respiration is stopped by cyanide, so they can move into the intestine by diffusion, which does not require energy.

---

---

(2)

## Lesson 7: Pathogens

**TASK 1:** Sort the diseases into the table. Is it a communicable or non-communicable disease?

|                                                                 |                                          |
|-----------------------------------------------------------------|------------------------------------------|
| Tuberculosis (air borne - bacteria)                             | Diabetes (pancreas malfunction)          |
| Respiratory Diseases (e.g. asthma)<br>(genetics/ environmental) | Cancer (genetics/ environmental)         |
| Arthritis (genetics/ environmental)                             | Cholera (water borne - bacteria)         |
| Measles (air borne - virus)                                     | Stroke (genetics/ environmental)         |
| Small pox (body fluids - virus)                                 | Chicken pox (air borne & fluids - virus) |
| High blood pressure( genetics/<br>environmental)                | Influenza (air borne - virus)            |
| Typhoid (water borne - bacteria)                                | Sickle cell anaemia (genetics)           |
| Heart Disease (genetics/<br>environmental)                      | HIV (blood and fluids - virus)           |

| Communicable                                                                                                                                                                                  | Non-communicable |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <p>Sort the diseases from the picture: communicable = caused by pathogens, spread between people (e.g. flu, cholera, malaria); non-communicable = not infectious (e.g. cancer, diabetes).</p> |                  |

## Questions

1. Describe how bacteria make people ill.

Bacteria make us ill by reproducing rapidly in the body and releasing toxins (poisons) that damage cells and tissues.

2. Give 2 examples of illnesses caused by fungi.

Athlete's foot and ringworm.....

3. Describe how viruses make people ill.

Viruses invade host cells and reproduce inside them, damaging or destroying the cells, which makes us ill.

4. Give 2 examples of illnesses caused by protists.

Malaria and amoebic dysentery.....

5. Explain why a virus is not a living organism.

Viruses are not made of cells and do not carry out the life processes (e.g. respiration, nutrition); they can only reproduce inside a living host cell.....

## Diseases and Their Causes

| Disease                     | Pathogen | How it's spread?                                  | Symptoms                                                                            | Treatment                                                          |
|-----------------------------|----------|---------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Flu</b>                  |          | Virus (influenza virus)                           | Droplet infection · inhaled droplets from coughs and sneezes                        | Fever, headache, aching muscles, cough, tiredness                  |
| <b>Pneumonia</b>            |          | Bacterium (e.g. <i>Streptococcus pneumoniae</i> ) | Droplet infection · inhaled droplets from coughs and sneezes                        | Fever, cough with phlegm, difficulty breathing, chest pain         |
| <b>Ringworm</b>             |          | Fungus                                            | Direct skin contact with infected person/animal or contaminated surfaces and spores | Itchy, red, ring-shaped rash on the skin or scalp                  |
| <b>Cholera</b>              |          | Bacterium ( <i>Vibrio cholerae</i> )              | Contaminated water or food (faecal-oral route)                                      | Severe watery diarrhoea and dehydration                            |
| <b>Tobacco Mosaic Virus</b> |          | Virus                                             | Contact with infected plants, contaminated tools or aphids                          | Mottled yellow patches on leaves, stunted growth                   |
| <b>Malaria</b>              |          | Protoctist ( <i>Plasmodium</i> )                  | Bite of the female <i>Anopheles</i> mosquito (vector)                               | Fever, shivering, headache, anaemia                                |
| <b>HIV (AIDS)</b>           |          | Virus                                             | Exchange of body fluids · sexual contact, infected blood, mother to baby            | Flu-like illness, weakened immune system, opportunistic infections |

### **Cells and Microbes – Summary**

- Living things are made from cells and carry out the processes of life
- There are 5 kingdoms – animals, plants, protoctists, fungi and bacteria
- Viruses are not living organisms as they do not have a cell surface membrane, cytoplasm or other organelles. They are parasites that must rely on a host to reproduce.
- Animal cells are different from plant cells as they do not have a cellulose cell wall, chloroplasts and a large permanent vacuole
- In a multicellular organism, there are differentiated cells each adapted for a specific function
- Stem cells are undifferentiated but with the right stimulus have the potential to become any specialized cell
- Molecules can move in and out of cells using the processes of diffusion, osmosis and active transport
- Diffusion is passive - molecules from an area of high concentration to an area of low concentration until dynamic equilibrium is reached.
- Osmosis is also passive but applies only to water, moving from an area of high water concentration (dilute solution) to an area of low water concentration (concentrated solution) through a selectively/partially permeable membrane
- Active transport requires energy and moves substances against the concentration gradient e.g. the movement of mineral ions into root hair cells from the soil
- The rate of diffusion is affected by: temperature, the surface area to volume ratio, the distance molecules have to travel and the difference in the concentration gradient
- Most multicellular organisms need to rely on a transport system to deliver substances to all cells as diffusion would not be efficient enough to support life
- Fungi can be unicellular like yeast or multicellular like mushrooms and mould
- All fungi are saprotrophs – they feed using extracellular digestion
- Fungal cells have a cell wall made from chitin, mycelium made of hyphae but cannot photosynthesize.
- Fungi have an important roles as a food source, in food production (bread and beer) and are essential for recycling nutrients (decomposers)
- Microbes that cause infectious diseases are called pathogens and may be a virus, bacterium, fungus or a protoctist.
- Bacteria have a cell wall, some can photosynthesize but they do not have a nucleus. Their DNA is circular and is found in the cytoplasm. They also contain plasmids.

**Unit 1 Cells and Microbes Revision Notes**

| Syllabus Statements                                                                                                                                                                                                                                                                                                                                                                                                                                       | Revision Notes                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1a) Characteristics of living organisms</b><br>1.1 Understand that living organisms share the following basic characteristics: <ul style="list-style-type: none"> <li>• they require nutrition</li> <li>• they respire</li> <li>• they excrete their waste</li> <li>• they respond to their surroundings</li> <li>• they move</li> <li>• they control their internal conditions</li> <li>• they reproduce</li> <li>• they grow and develop.</li> </ul> | <p>Living organisms share these characteristics: they require nutrition, respire, excrete waste, respond to their surroundings, move, control internal conditions, reproduce and grow and develop.</p>                                                                              |
| <b>1b) Variety of living organisms</b><br>1.2 describe the common features shown by eukaryotic organisms: plants, animals, fungi and protocists<br><br>1.3 describe the common features shown by prokaryotic organisms such as<br><br>1.4 Recall the term 'pathogen' and know that pathogens may be fungi, bacteria, protocists or viruses.                                                                                                               | <p>Plants: cellulose wall, chloroplasts, large vacuole. Animals: no cell wall, multicellular, move. Fungi: chitin wall, saprophytic, hyphae. Protocists: single-celled, e.g. Amoeba, Euglena.</p> <p>Animals:</p> <p>Fungi:</p> <p>Protocists:</p> <p>Bacteria:</p> <p>Viruses:</p> |
| <b>2a) Level of organisation</b><br>2.1 describe the levels of organisation within organisms: organelles, cells, tissues, organs and systems.                                                                                                                                                                                                                                                                                                             | <p>Organelles form cells, cells form tissues, tissues form organs, organs form systems: e.g. nucleus · muscle cell · muscle tissue · stomach · digestive system.</p>                                                                                                                |
| <b>2b) Cell structure</b><br>2.2 describe cell structures, including the nucleus, cytoplasm, cell membrane, cell wall, mitochondria, chloroplasts, ribosomes and vacuole<br><br>2.3 describe the functions of the nucleus, cytoplasm, cell membrane, cell wall, mitochondria, chloroplasts, ribosomes and vacuole<br><br>2.4 know the similarities and differences in the structure of plant and animal cells.                                            | <p>Nucleus: DNA/control. Cytoplasm: reactions. Membrane: entry/exit. Cell wall: support. Mitochondria: respiration. Chloroplasts: photosynthesis. Ribosomes: protein synthesis. Vacuole: cell sap.</p>                                                                              |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>2d) Movement of substances into and out of cells</b></p> <p>2.15 understand the processes of diffusion, osmosis and active transport by which substances move into and out of cells</p> <p>2.16 understand how factors affect the rate of movement of substances into and out of cells, including the effects of surface area to volume ratio, distance, temperature and concentration gradient</p> <p>2.17 Practical: investigate diffusion and osmosis using living and no-living systems</p> <p>2.51 understand why simple, unicellular organisms can rely on diffusion for movement of substances in and out of the cell</p> | <p>Diffusion: down a concentration gradient. Osmosis: water through a partially permeable membrane. Active transport: against the gradient, uses energy. Rate: SA:vol, distance, temperature, gradient.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Revision - Knowledge Recall Multiple Choice Test

1. Which of the following sets of conditions would make diffusion of a substance into a cell fastest?

Fastest diffusion occurs when (big) difference in concentration and a high temperature.

|                                    | cell surface area | concentration difference |
|------------------------------------|-------------------|--------------------------|
| <input checked="" type="radio"/> A | high              | high                     |
| B                                  | high              | low                      |
| C                                  | low               | high                     |
| D                                  | low               | low                      |

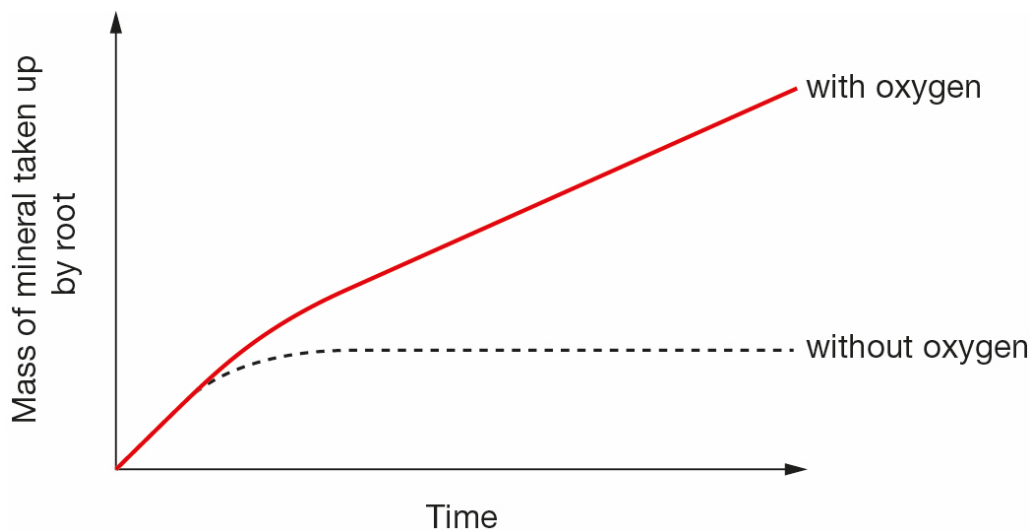
2. Here are some statements about active transport.

1. active transport pumps particles down a concentration gradient
2. active transport requires ATP Correct - active transport requires ATP (energy released by respiration).
3. active transport requires living cells Correct - active transport requires living cells, because only living cells can supply the ATP.

Which of the statements are correct? Statements 2 and 3 are correct (1 is wrong because the particles are moved against the concentration gradient).

- A 1 and 2  
 B 1 and 3  
☒ C 2 and 3  
 D 1, 2 and 3

3. The graph shows the transport of mineral ions into a plant root hair cell over time with oxygen and without oxygen.



Which processes does the root hair cell use to transport the mineral ions?

- A active transport alone  
☒ B active transport and diffusion  
 C diffusion and osmosis  
 D diffusion alone

4. Which of the following is the correct order of structures? Cell, then tissue, then organ, then organ system (smallest structure first).

- A cell, tissue, system, organ  
☒ B cell, tissue, organ, system  
 C cell, organ, system, tissue  
 D cell, system, tissue, organ

Incorrect - active transport pumps particles AGAINST the concentration gradient, not down it.

Both diffusion and active transport - diffusion continues without oxygen, but the extra uptake when oxygen is present is active transport using ATP.

5. Which row correctly states the functions of the organelles?

~~Mitochondria carry out photosynthesis (they consume CO<sub>2</sub>); chloroplasts carry out protein synthesis;~~

|          | <b>mitochondria</b> | <b>ribosomes</b>  | <b>chloroplasts</b> |
|----------|---------------------|-------------------|---------------------|
| A        | photosynthesis      | protein synthesis | respiration         |
| B        | photosynthesis      | respiration       | protein synthesis   |
| C        | respiration         | photosynthesis    | protein synthesis   |
| <b>D</b> | respiration         | protein synthesis | photosynthesis      |

6. Which of these structures are found in both plant and animal cells?

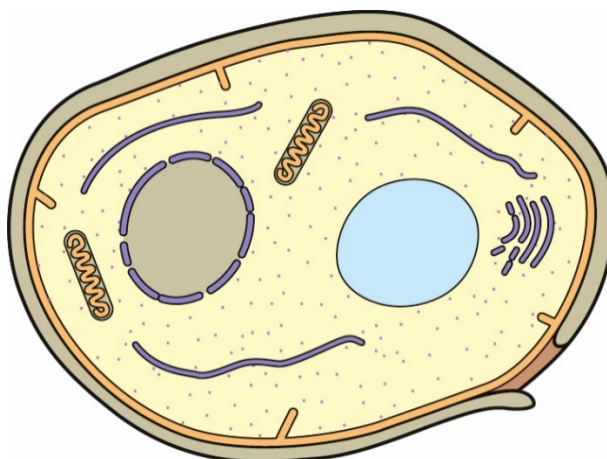
- A cell membrane, cell wall and large vacuole  
**B** cell membrane only  
 C cell wall and large vacuole  
 D cell wall only

7. Which row correctly identifies a prokaryotic cell?

~~Not all cells have chloroplasts or mitochondria and ribosomes are in~~

|          | <b>nucleus</b> | <b>mitochondria</b> |
|----------|----------------|---------------------|
| <b>A</b> | absent         | absent              |
| B        | absent         | present             |
| C        | present        | absent              |
| D        | present        | present             |

8. The diagram shows a yeast cell.

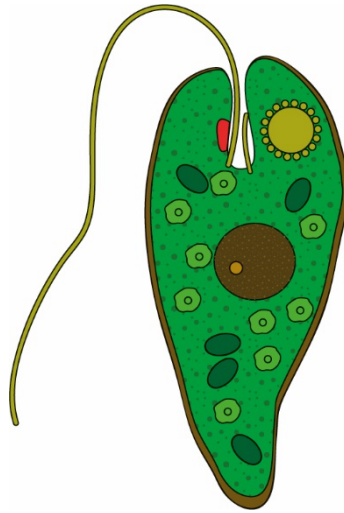


9. Which of the following statements about yeast is correct?

- A it is a eukaryote and has a cell wall made of cellulose  
**B** it is a eukaryote and has a cell wall made of chitin  
 C it is a prokaryote and has a cell wall made of cellulose  
 D it is a prokaryote and has a cell wall made of chitin

· Yeast cell diagram: complete the labels, e.g. cell wall (chitin), cell membrane, cytoplasm, nucleus, vacuole.

10. The diagram shows a single celled organism called *Euglena*.



In which kingdom should *Euglena* be placed?

- A animals
- B fungi
- C plants
- ☒ D prototists

11. Which of the following can have both multicellular and single-celled organisms?

- A animals
- B bacteria
- ☒ C fungi
- D plants

12. Which of the following rows matches the carbohydrate molecule with a correct group?

|                                    |         |                 |                                                          |
|------------------------------------|---------|-----------------|----------------------------------------------------------|
|                                    |         | <b>glycogen</b> | <b>starch</b>                                            |
|                                    |         |                 | <i>Glycogen is the storage carbohydrate in animals,</i>  |
| A                                  | animals | animals         | <i>starch is the storage carbohydrate in fungi</i>       |
| B                                  | fungi   | fungi           | <i>starch is the storage carbohydrate in plants, and</i> |
| C                                  | plants  | plants          | <i>cellulose found in plant cell walls.</i>              |
| <input checked="" type="radio"/> D | plants  | plants          | <i>cellulose found in animal cell walls.</i>             |

13. Below is a list of the characteristics of life.

movement  
respiration  
sensitivity  
growth  
reproduction  
excretion  
nutrition  
control

How many of these characteristics of life do bacteria exhibit?

- A 5
- B 6
- C 7
- ☒ D 8

Kingdom Protocista (prototists) - *Euglena* is a single-celled organism showing features of both plants and animals.

All 7 of them - bacteria show movement, respiration, sensitivity, growth, reproduction, excretion and nutrition.

14. Which of the following statements about viruses is correct?

- A viruses are made of cells
- B viruses can reproduce outside the body
- ☒ C viruses carry DNA or RNA
- D viruses release energy by respiration

15. Which of the following groups contain examples of pathogens?

|                                    | <b>bacteria</b> | <b>protists</b> | <b>viruses</b> |
|------------------------------------|-----------------|-----------------|----------------|
| A                                  | no              | no              | yes            |
| B                                  | no              | yes             | no             |
| <input checked="" type="radio"/> C | yes             | yes             | yes            |
| D                                  | yes             | no              | yes            |

16. Below are some statements about fungi.

- fungal hyphae release enzymes to digest food
- the network of hyphae is called a mycelium
- many fungi use saprophytic nutrition
- examples of fungi include yeast and *Mucor*

How many of the statements are correct?

- A 1
- B 2
- C 3
- ☒ D 4

All of the statements are correct - fungi have nuclei, cell walls of chitin, feed saprotrophically and include yeast and *Mucor*.

**Revision Exam: Application of Knowledge**

**1 a)** Although plants and animals have many different features, they also have some features in common.

The table lists some features.

In each box, place a tick(✓) if the feature is present or a cross (x) if the feature is absent. One has been done for you. (4)

| Feature                                             | Plants | Animals |
|-----------------------------------------------------|--------|---------|
| Can move from place to place                        | x      | ✓       |
| Can carry out photosynthesis Plants: · Animals: x   |        |         |
| Are multicellular Plants: · Animals: ·              |        |         |
| Have cells with cell walls Plants: · Animals: x     |        |         |
| Store carbohydrate as glycogen Plants: x Animals: · |        |         |

**(b)** Organisms that cause disease are known as pathogens.

Give two groups of organism that include pathogens.

(2)

Bacteria.....

Viruses.....

(Total for Question 1= 6 marks)

**2.** The table lists the names of different structures found within a human.

Complete the table by numbering each structure in order of its size.

Use number 1 for the smallest structure through to number 4 for the largest.

| Name of structure | Order of size |
|-------------------|---------------|
| Brain 3           |               |
| Nucleus 1         |               |
| Nervous system 4  |               |
| Nerve cell 2      |               |

(Total 3 marks)

3. John wanted to investigate the effect of the size of potato tissue on the rate of osmosis.

He cut three different sized cubes of potato, one  $0.5 \times 0.5 \times 0.5$  cm, one  $1 \times 1 \times 1$  cm and one  $2 \times 2 \times 2$  cm.

He weighed the potato cubes and recorded their masses.

He then placed each cube into a beaker of distilled water and left them for 1 hour.

He weighed them again and recorded their new masses.

In each case the mass of the potato cubes increased.

The table shows his results.

| Potato cube size in cm      | Original mass in g | Final mass in g | Percentage change in mass (%) |
|-----------------------------|--------------------|-----------------|-------------------------------|
| $0.5 \times 0.5 \times 0.5$ | 0.06               | 0.07            | 16.67                         |
| $1 \times 1 \times 1$       | 0.51               | 0.56            |                               |
| $2 \times 2 \times 2$       | 4.04               | 4.10            | 1.49                          |

His teacher told him to calculate the percentage change in mass of each cube.

(a) (i) Calculate the percentage change in mass for the  $1 \times 1 \times 1$  cm cube.  
Show your working.

(2)

Answer:..... Percentage change in mass =  $\frac{(\text{change in mass} \div \text{starting mass}) \times 100}{\text{calculated from the results in the table.}}$

(ii) Explain why John converted change in mass to percentage change in mass. (1)

The cubes had different starting masses, so converting to percentage change allows a fair comparison .....  
between them.....  
.....

(b) Use your knowledge of osmosis to explain why the mass of each cube increased. (3)

The water was more dilute than the cell sap, so water moved into the potato cubes by osmosis through the partially permeable membrane, from high to low water concentration, increasing their mass.....  
.....  
.....  
.....

(c) John's teacher suggested that the different sized cubes have a different surface area and volume.

Use the formulae below to calculate the surface area (SA) of each cube, the volume (Vol) of each cube and their SA:Vol ratios.

$$SA = 6 \times (\text{cube side in cm})^2$$

$$Vol = (\text{cube side in cm})^3$$

Write your answers in the table below. One cube has been done for you. (3)

| Cube side in cm | Surface area in cm <sup>2</sup>                                      | Volume in cm <sup>3</sup> | Sa:Vol ratio |
|-----------------|----------------------------------------------------------------------|---------------------------|--------------|
| 0.5             | 1.5                                                                  | 0.125                     | 12           |
| 1.0             | SA = 6 cm <sup>2</sup> ; Vol = 1 cm <sup>3</sup> ; SA:Vol = 6:1 (6)  |                           |              |
| 2.0             | SA = 24 cm <sup>2</sup> ; Vol = 8 cm <sup>3</sup> ; SA:Vol = 3:1 (3) |                           |              |

(d) Explain the effect of the different SA:Vol ratios on the rate of osmosis into the potato. (2)

The smaller cube has a larger surface area to volume ratio, so there is relatively more membrane surface.. for water to enter, so osmosis (water uptake) into it is faster than into the larger cube.....

.....

.....

.....

(e) Potato tissue is made from plant cells.

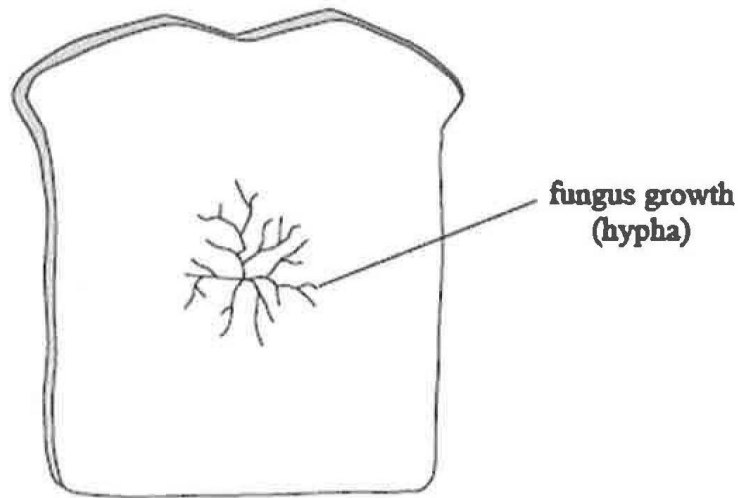
Draw and label a plant cell in the space below.

(3)

· Draw and label a plant cell: cell wall, cell membrane, cytoplasm, nucleus, chloroplasts, large permanent vacuole, mitochondria, ribosomes.

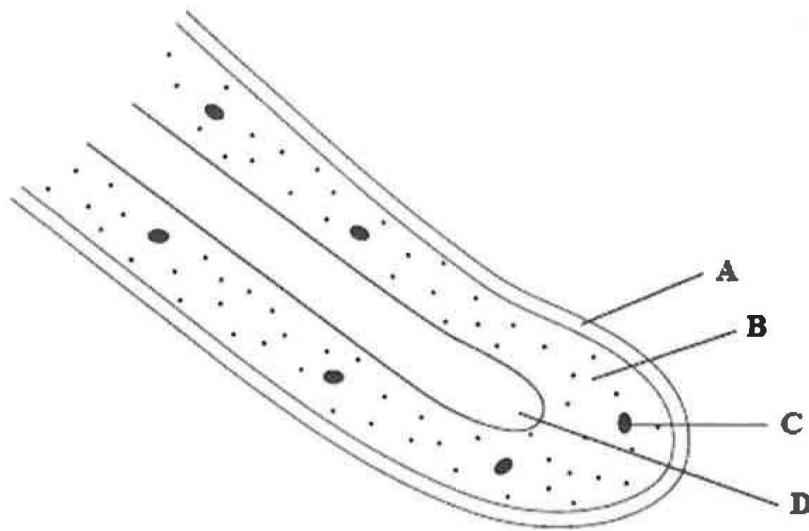
(Total for Question 3 =11 marks)

4. The drawing shows a piece of bread. The bread is going mouldy because a fungus is growing and feeding on it. When the fungus grows it produces many threads (hyphae) that spread over the bread.



The hyphae secrete enzymes that digest the bread. The products from this digestion are then absorbed by the fungus.

- (a) The diagram shows a magnified view of a single hypha from the fungus.



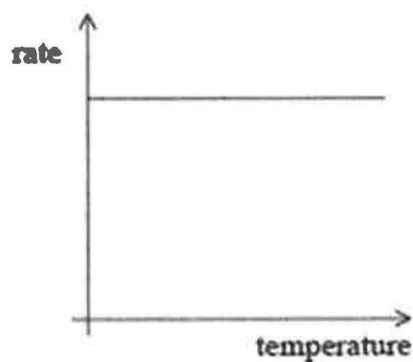
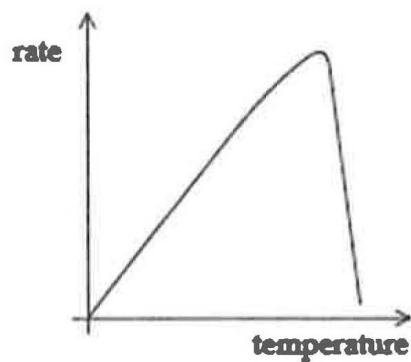
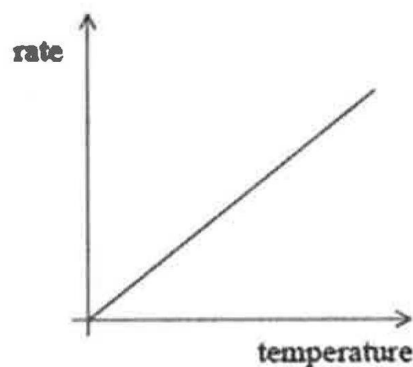
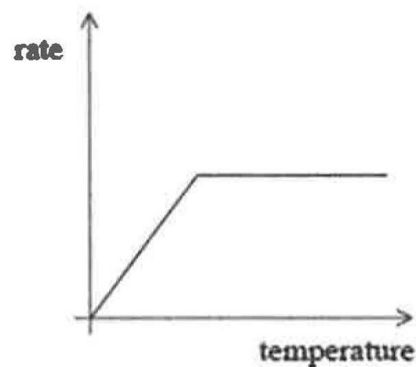
- (i) Name the parts labelled A, B, C and D.

A: .....  
B: .....  
C: .....  
D: .....

(Total for Question 4 = 4 marks)

· Name the parts labelled A-D on the diagram; if it is a cell diagram, typical answers are A nucleus, B cytoplasm, C cell membrane, D mitochondria (for a plant cell: chloroplast/cell wall/vacuole).

5. The graphs show the effect of changing the temperature on the rate of movement of substances into cells.

**A****B****C****D**

- (i) Which graph shows the effect of temperature on diffusion? (1)

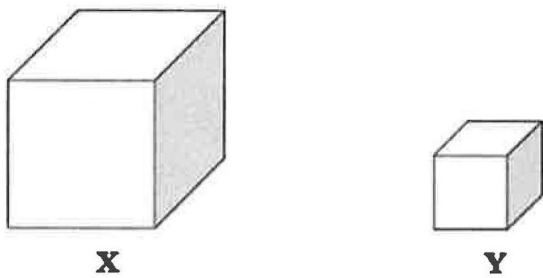
The graph showing rate of diffusion increasing steadily as temperature increases. (particles gain kinetic energy and move faster).....

- (ii) Which graph shows the effect of temperature on active transport? (1)

The graph showing rate rising to an optimum temperature then falling as carrier proteins/enzymes denature.....

(Total for Question 5= 2 marks)

6. The diagram shows two different-sized potato cubes, X and Y, that were placed in distilled water for one hour.



(a) Explain why both cubes gained in mass after one hour. (3)

.....

.....

.....

.....

(b) (i) Put a tick ✓ in the row of the table that correctly describes the change in mass for cube X compared to cube Y.

Tick the correct row for each cube using the

| Mass of water absorbed in g<br><small>results: the cube with the larger surface area</small> | % increase in mass | tick |
|----------------------------------------------------------------------------------------------|--------------------|------|
| more<br><small>to volume ratio absorbs more water and has</small>                            | lower              |      |
| less<br><small>a higher % increase in mass.</small>                                          | lower              |      |
| less                                                                                         | higher             |      |
| more                                                                                         | higher             |      |

(ii) Calculate the percentage increase in length of the potato cylinders after one hour in water. Show your working. (2)

~~Percentage increase in length = (increase in length ÷ original length) × 100, using the data from the results.~~

Answer: ~~(increase in length ÷ original length) × 100 (%)~~

(iii) Explain why the potato cylinders increased in length when they were in water. (3)

.....

.....

.....

.....

c) Suggest why the graph levelled out for solutions of sucrose greater than 5 arbitrary units. (1)

.....

The solutions were more dilute than the cell sap, so water moved into the potato cells by osmosis through the partially permeable membrane, from high to low water concentration, so mass increased.

Water entered the potato cells by osmosis (the solution was more dilute than the cell sap), so the cells swelled and became turgid, increasing the length of the cylinders.

Above 5 units the cells had lost as much water as they could (fully plasmolysed), so further increases in sucrose concentration caused no further decrease in length.

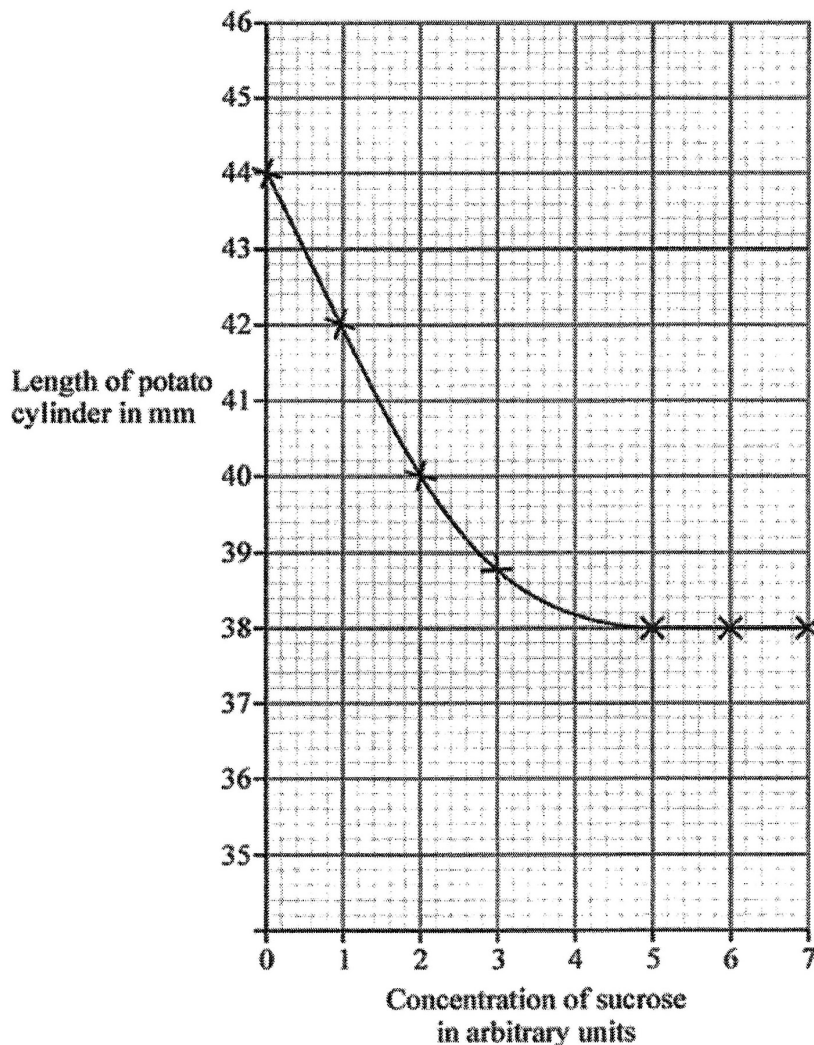
7. A group of students investigated osmosis.

They cut potato cylinders, 40 mm in length.

Some of these cylinders were put into water and some into sucrose solutions of different concentration.

After one hour the length of each potato cylinder was measured.

The graph shows the length of the cylinders after one hour plotted against the concentration of sucrose.



Use the graph and your knowledge to answer the questions.

- (a) (i) which concentration of sucrose resulted in exactly 2 mm increase in length of potato cylinders? (1)

Read the sucrose concentration from the graph where the increase in length is 2 mm (where the curve crosses 2 mm on the y-axis).....

- (ii) Which solution of sucrose had the same concentration as the cell sap in the potato cells? Give a reason for your answer. (2)

The sucrose concentration where the graph shows zero change in length - no net water movement means the solution has the same concentration as the cell sap (isotonic), so no osmosis occurs.

## Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p11: Animal cell labels: cell membrane, cytoplasm, nucleus, mitochondria, ribosomes. Plant cell labels: cell wall, cell membrane, cytoplasm, nucleus, chloroplast, large vacuole, mitochondria, ribosomes.

p12: Microscope labels: eyepiece (top of the tube), objective lens (on the nosepiece above the stage), stage, slide (on the stage), light/lamp (in the base), coarse focus (large wheel) and fine focus (small wheel).

p30: Graph: concentration of sugar solution on the x-axis, % change in mass on the y-axis; plot the points and draw a smooth line of best fit. The concentration where the line crosses zero is the concentration inside the potato cells.

p39: Treatment: Flu - rest/vaccine; Pneumonia - antibiotics; Ringworm - antifungal cream; Cholera - oral rehydration salts; TMV - no cure (remove plants); Malaria - antimalarials; HIV - antiretrovirals.